Exact Compositional Transfer of Bounded Linear Recovery

Tavis Rudd

August 2026

Abstract

In a concatenated linear code, an outer equation selects which inner functional each block must realize. Recording only the least helper count forgets that label and can give the wrong finite threshold. We first show that, within one inner block, relative generalized Hamming weights of the canonical shortening–puncturing pair are exactly the minimum helper unions for recovering each target dimension. For concatenation, we determine the minimum helper cost of a recovery leaving the target block. The answer is a labelled min–sum formula over functionals compatible with the outer code. Its local costs are minimum prescribed-coset supports under target normalization. These labelled costs compose associatively through finite concatenation towers. Sufficient outer dual distance excludes the nonzero functional terms and reduces the answer to $M_t(D_P, K_P) + d(I^\perp)$. Below the resulting helper-cost threshold, restriction and zero-extension preserve every normalized recovery equation and its exact helper support, not only the minimum cost. Thus relative weights solve the single-block minimum problem, while labelled coset costs provide the finer data needed for exact composition. A companion exact implementation evaluates the recursion and returns optimizing coefficient witnesses; no proof depends on software.

1 Introduction

Which recovery information survives concatenation? A scalar locality or confinement threshold does not: the outer code can force an intermediate functional whose cost was discarded when the scalar minimum was taken. The problem is therefore compositional. We seek finite data that determine exactly when a recovery equation can leave its inner block and that can be propagated through another concatenation level.

A linear recovery equation expresses a target combination as a linear combination of helper symbols; its cost is the number of distinct helpers used. It is *confined* when every coefficient lies in the target's own inner block and *nonconfined* when at least one helper lies in another block.

The information loss can be seen before any coding notation. Suppose an inner block can realize two intermediate functionals a and b at respective costs one and two. The scalar minimum is one. If an outer equation forces label b, however, the relevant cost is two; replacing the labelled table $(a \mapsto 1, b \mapsto 2)$ by its minimum has destroyed the answer. The exact theory below makes this schematic example intrinsic and, in Proposition 4.2, realizes it while holding the unlabelled recovery data fixed.

Here exact transfer means more than equality of a numerical parameter. Below a helper radius, every global recovery is uniquely the zero-extension of a recovery inside the target block. Formally, restriction to that block and zero-extension back into the concatenation are inverse bijections on normalized recovery-equation systems, preserving coefficients and exact helper supports. Normalization means that the target coefficients act as the identity on the recovered subspace. This is the structure needed to transport recovery-set overlap, bounded reliability, and capacity constraints.

Three projections of the recovery data answer different questions. Relative generalized Hamming weights give the minimum helper union at each recovered dimension. Exact helper-support families retain the overlap data needed for reliability. Labelled prescribed-coset support functions start from the same equations but fix the induced intermediate functional before minimizing, which keeps the labels needed for concatenation. Figure 1 shows how these projections meet; neither the support-family branch nor the labelled-cost branch contains the other.

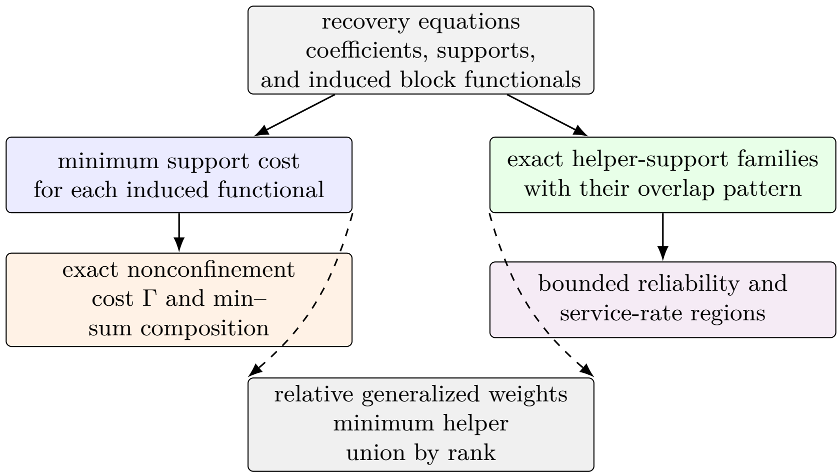

Figure 1: Two projections of coefficient-labelled recovery equations answer different questions. On the left, minimizing within each induced-functional class gives the labelled costs used for exact numerical composition; a solver must also retain a minimizing equation to propagate coefficient witnesses. Exact support families retain the overlaps needed for stochastic and fractional recovery questions. Relative weights are the common minimum-cost projection (dashed arrows), complete for helper-union minima but not for either richer question.

Let q be a prime power, let $I \leq \mathbb{F}_q^E$, and partition the coordinates as $E = P \sqcup J$ into targets and helpers. The associated nested code pair is

$$D_P = \operatorname{punct}_J(I^\perp), \qquad K_P = \operatorname{short}_J(I^\perp).$$

Here punct_J restricts a word to the helper coordinates J, whereas short_J first requires the target coordinates P to vanish and then restricts to J. For a full-row-rank encoder $G = (G_P \mid G_J)$, put $W_P = \operatorname{im} G_P \cap \operatorname{im} G_J$. The exact sequence in Proposition 3.1 identifies $D_P/K_P \cong W_P$, the space of target combinations recoverable from the helpers. We assume $0 < \dim I < |E|$, so both the represented message space and $I^\perp$ are nonzero. We use the convention $d(0) = \infty$ for the zero code. Here t is the dimension of a recovered subspace of W_P, not the number of erased coordinate positions. Recovering all values on P is the special case $\operatorname{im} G_P \subseteq \operatorname{im} G_J$ and $t = \dim \operatorname{im} G_P$; dependent target columns may make this dimension smaller than $|P|$.

The first layer is classical nested-code support theory in recovery language: the minimum helper union for recovered dimension t is $M_t(D_P, K_P)$, the tth relative generalized Hamming weight of the pair. Theorem 3.3 gives the exact support-preserving correspondence. The principal result begins at the next layer, where the outer code can distinguish functional labels having the same minimum support.

For concatenation, identify the message space of I with a finite extension field $L/\mathbb{F}_q$ and retain the chosen encoder $L \to I$. We call this encoder together with its image a *represented inner code*; the representation matters because it assigns field functionals to coefficient fibres.

For an outer code O, let $\mathrm{FD}(O)$ be its functional dual: the tuples of $\mathbb{F}_q$-linear functionals on L whose coordinatewise evaluations sum to zero on every word of O. For a fixed target block j and a nonzero recovered subspace $T \leq W_P$, put

$$\mathcal{B}_T(O) = \mathrm{Hom}_{\mathbb{F}_q}\big(T, \mathrm{FD}(O)\big).$$

Writing $B = (B_h)_{h=1}^N$, let $\mu_{I,P,T}(B_j)$ be the least helper support in the target block among systems normalized on T and inducing B_j, and let $\lambda_{I,T}(B_h)$ be the corresponding least support without target normalization in block h. Section 4 defines these quantities precisely and proves

$$\Gamma_{j,T}(O,I) = \min\left\{ \mu_{I,P,T}(0) + d(I^\perp),\ \min_{0 \neq B \in \mathcal{B}_T(O)} \left(\mu_{I,P,T}(B_j) + \sum_{h \neq j} \lambda_{I,T}(B_h) \right) \right\}.$$

The two terms exhaust the possibilities. A zero outer-functional label gives one inner recovery system plus a minimum inner-dual perturbation in another block. A nonzero label prescribes one coset in every selected block. Put $\Gamma_{j,t}(O,I) = \min\{\Gamma_{j,T}(O,I) : T \leq W_P,\ \dim T = t\}$.

Theorem 1.1 (Main characterization: exact finite transfer)**.** *Let $\ell = \dim W_P \geq 1$ and $1 \leq t \leq \ell$. Let I be used as the represented inner code in a concatenation with an L-linear outer code $O \leq L^N$ with $N \geq 2$, where L is the message space of I, and fix a block j onto which O projects nontrivially. The minimum cost of a nonconfined recovery system—one using at least one helper outside block j—in recovered dimension t is $\Gamma_{j,t}(O,I)$. Hence confinement through radius r holds exactly when $r < \Gamma_{j,t}(O,I)$.*

The content is the correspondence behind the formula: every global recovery system determines exactly one compatible labelled family in one of its two terms, and every such family lifts to a global recovery with the displayed sum of block supports.

Outer-distance collapse. Sufficient outer dual distance rules out every competing nonzero-label sector, leaving the relative-weight cost plus one inner-dual perturbation. The exact characterization gives

$$d(O^\perp) \geq M_t(D_P, K_P) + d(I^\perp) + 1 \quad \Longrightarrow \quad \Gamma_{j,t}(O,I) = M_t(D_P, K_P) + d(I^\perp).$$

For a fixed radius r, the weaker condition $d(O^\perp) > r + 1$ reduces the exact criterion to

$$r < M_t(D_P, K_P) + d(I^\perp).$$

When this outer-distance condition and the displayed inequality both hold, zero-extension gives a bijection on the normalized recovery-equation systems and their exact helper supports. Consequently, for any outer family with dual distance tending to infinity, the same statement holds for all sufficiently long concatenations at every fixed r.

The second central result is the closure law in Theorem 4.3: the labelled prescribed-coset functions entering $\Gamma_{j,T}$ compose associatively by exact min–sum substitution. The numerical functions retain the minimum in each labelled fibre. If the recursion also stores a minimizing lift in each attained fibre, those minimizers propagate an optimizing coefficient-level witness through the tower. Transporting the full family instead requires the full lift sets used in the proof of the closure theorem.

Thus three outputs must not be conflated: a fibre minimum is one number, a stored minimizer is one recoverable witness, and the full lift set is the complete family of coefficient-labelled systems and supports.

The rank-one consequence is especially simple: a nonconfined recovery on a higher-dimensional target restricts to a nonconfined recovery on a target line without increasing its helper support. Hence confinement through a fixed radius holds at every recoverable rank if and only if it holds at rank one (Corollary 4.6). Theorem 4.4 separately identifies which numerical label distinctions compatible outer codes can detect, and Proposition 4.5 shows that target rank t needs no test whose outer functional dual has dimension greater than t.

The label loss is already visible over $\mathbb{F}_2 \subseteq \mathbb{F}_4$. Two three-column encoders have the same binary image code, dual, abstract nested pair, complete relative-weight hierarchy, dual distance, and unlabelled helper-support family. Their functional labels differ, and one fixed outer functional-dual line gives exact nonconfinement costs one and two. Proposition 4.2 gives the calculation before the closure theorem. Thus every unlabelled and scalar explanation is held fixed; only alignment with the intermediate functional labels changes.

The restriction–extension bijection yields the later transfer results. It preserves every block-local statistic determined by normalized recovery systems of cost at most r. After forgetting coefficients, it also preserves statistics determined by inclusion-minimal helper supports, including bounded repair reliability and the bounded service-rate region. This statement is block-local: the upward-closed recovery-set family in the full concatenated coordinate set also contains irrelevant cross-block supersets. For example, a copied minimal support inside block j may be enlarged by an unused helper in another block; that larger recovery set changes neither the minimal support nor any bounded success event containing it.

The theorem contains the previous one-coordinate criterion. If $P = \{x\}$ and $t = 1$, then $M_1(D_P, K_P)$ is the least number of helpers in a dual word through x, and

$$z_x(I) = M_1(D_P, K_P) + 1 + d(I^\perp).$$

Here $z_x(I)$ is the corresponding total-word-weight threshold: the target coordinate contributes one in addition to the helper cost. Thus $r < M_1 + d(I^\perp)$ is exactly $r + 1 < z_x(I)$.

Relative generalized Hamming weights (RGHWs) and relative dimension/length profiles (RDLPs) already describe support and access thresholds for nested codes in wiretap and ramp-secret-sharing settings [26, 22, 14, 37, 13]. The identity between the minimum helper-union quantity $\mu_t(P)$ defined below and $M_t(D_P, K_P)$ is this standard invariant specialized to the canonical target/helper pair. Recovery-set and locality formulations retain helper supports but generally discard the normalized coefficients and functional labels that couple concatenation blocks [36, 15, 33, 1, 28, 16]. Work on recovering entire subspaces from simplex-code columns optimizes disjoint support families, not their coefficient-labelled transfer through concatenation [7]. Regenerating-code models optimize download and subpacketization rather than the scalar helper-support cost used here [38, 41, 18, 29].

Service-rate regions fractionally allocate requests among recovery sets under server-capacity constraints [3, 4]. Our consequence transfers the radius-bounded region generated by the preserved helper supports; it does not introduce or characterize the general service-rate polytope. Likewise, after choosing a basis of T, the ordinary local cost in Γ is the fixed-instance minimum union support of representatives of prescribed cosets, as in generalized coset weights; generalized covering radii maximize the corresponding quantity over coset tuples [42, 10]. The additional data here are target normalization and the labels imposed by the outer functional dual.

Concatenation has long been used to construct locally repairable codes (LRCs) with prescribed parameter properties [24, 12, 20]. Matrix-product constructions can transport recovery structures from constituent codes or from the defining matrix, while Jin and Fu fix a binary $[3, 2, 2]$ inner code and choose $\mathbb{F}_4$ outer codes to obtain parameter-optimal binary locality-two codes. These works give sufficient recovery constructions and parameter consequences. Here the inner code and target split are arbitrary, and the question is the exact obstruction to preserving every bounded dual recovery equation. The finite formula treats each recovered dimension separately and needs no outer-distance hypothesis; under the stated outer-distance condition, it reduces to the RGHW formula. We make no minimum-bandwidth claim.

The functional-dual decomposition is classical concatenation machinery, and min–sum elimination also appears in soft and generalized-concatenated decoding [11, 8, 2, 17, 5]. Label-refined precedents are also important: soft concatenated decoding passes symbol-indexed reliabilities, and input–output weight enumerators can depend on the chosen generator presentation and compose by convolution in recursive constructions [17, 25]. Nogin's message weight functions provide a related representation-sensitive refinement of ordinary weight data [32]. These precedents retain labels for decoding or enumeration; they do not impose target normalization or identify the exact first nonconfined support. Here target-normalized prescribed-coset costs characterize that support and form a closed state under repeated concatenation.

The terminology "contextual equivalence" and "congruence" describes the response to all permitted outer compositions. It parallels behavioural minimization for weighted automata over semirings [31], but we make no new general automata-minimization claim: Theorem 4.4 identifies the separating outer-code probes for this finite coding problem.

ergodis (Exact Recovery, Global Optimization, and Invariant Synthesis) is an exact compiler and solver derived from these reductions. From finite linear recovery data and optional capacity constraints it composes labelled costs, computes confinement thresholds and schedules, and returns replayable coefficient-level witnesses. Section 6 gives correctness and complexity bounds and evaluates representative storage, graph, code-design, and tower applications against matched exact controls. No mathematical result depends on the implementation or its measurements.

2 Recovery sets and normalized equations

Let $C \le \mathbb{F}_q^E$ be a linear code. We first separate four layers that are often harmlessly conflated when only locality is at issue.

Fix a target $x \in E$ and a helper bound r. A dual word $w \in C^\perp$ with $w_x \neq 0$ gives

$$c_x = \sum_{y \neq x} -\frac{w_y}{w_x} c_y \qquad (c \in C).$$

Its exact helper support is $\mathrm{supp}(w) \setminus \{x\}$. Define

$$\mathcal{H}_x^{(\le r)}(C) = \{\mathrm{supp}(w) \setminus \{x\} : w \in C^\perp, w_x \neq 0, \mathrm{wt}(w) \le r+1\}.$$

The usual bounded recovery-set family is the upward closure of $\mathcal{H}_x^{(\le r)}(C)$ inside the subsets of $E \setminus \{x\}$ of size at most r. Its inclusion-minimal members form the minimal recovery-set clutter. Here a clutter is a family of sets none of which contains another. Thus exact supports, recovery sets, and minimal recovery sets are distinct but determine the same monotone success event.

The coefficient layer is

$$\mathcal{P}_x^{(\le r)}(C) = \{w \in C^\perp : w_x = 1, \mathrm{wt}(w) \le r+1\}.$$

These normalized recovery equations determine their exact supports, while the converse generally fails. The operational model is scalar exact repair: each participating helper supplies one symbol of

$\mathbb{F}_q$. Array-code access, subpacketization, and repair bandwidth are different optimization problems [38, 41, 18, 29].

If $A \subseteq E \setminus \{x\}$ is the surviving helper set, put

$$f_{x,r}(A) = 1 \quad \Longleftrightarrow \quad A \text{ contains a member of } \mathcal{H}_x^{(\leq r)}(C).$$

Under independent helper survival with probabilities (s_y), the bounded repair reliability is

$$R_{x,r}((s_y)) = \mathbb{E}[f_{x,r}].$$

It depends only on the minimal recovery sets, but it is not determined by their minimum cardinality. Section 5 shows that even the complete relative-weight hierarchy does not determine the corresponding bounded reliability law.

For several target coordinates, a recovery system is a linear family whose image is a dual subspace rather than one dual word. Let $P \subseteq E$, $J = E \setminus P$, and let $G = (G_P \mid G_J)$ be a full-row-rank generator matrix. We view its blocks as maps $G_P : \mathbb{F}_q^P \to \mathbb{F}_q^k$ and $G_J : \mathbb{F}_q^J \to \mathbb{F}_q^k$. If $T \leq \operatorname{im} G_P$ is a subspace of target linear combinations, a normalized recovery system for T consists of linear maps

$$\alpha : T \longrightarrow \mathbb{F}_q^P, \qquad \beta : T \longrightarrow \mathbb{F}_q^J$$

such that

$$G_P\alpha = \mathrm{id}_T, \qquad G_J\beta = -\,\mathrm{id}_T\,.$$

For each $t \in T$, the vector $(\alpha(t), \beta(t))$ is then a dual word. The map α records the chosen target normalization and β records the helper coefficients.

The helper union of the system is the support of $\beta(T)$:

$$\operatorname{supp}(\beta(T)) = \{j \in J : \text{some } t \in T \text{ has } \beta(t)_j \neq 0\}.$$

Its helper cost is $|\operatorname{supp}(\beta(T))|$. We call a target subspace helper- recoverable when it lies in

$$W_P = \operatorname{im} G_P \cap \operatorname{im} G_J.$$

Only this recovered subspace dimension is counted below. Relations supported entirely on P have recovered dimension zero and do not consume helpers.

3 Puncturing, shortening, and relative recovery costs

Continue with $G = (G_P \mid G_J)$, viewed as $G_P : \mathbb{F}_q^P \to \mathbb{F}_q^k$ and $G_J : \mathbb{F}_q^J \to \mathbb{F}_q^k$, and set

$$U_P = \operatorname{im} G_P, \qquad W_P = U_P \cap \operatorname{im} G_J,$$

$$K_P = \ker G_J, \qquad D_P = G_J^{-1}(U_P).$$

Proposition 3.1 (Associated nested code pair)**.** *Restriction of G_J gives an exact sequence*

$$0 \longrightarrow K_P \longrightarrow D_P \xrightarrow{G_J} W_P \longrightarrow 0. \tag{1}$$

Proof. By definition, $K_P = \ker G_J$ is contained in $D_P = G_J^{-1}(U_P)$. The kernel of the restricted map is therefore K_P. Its image is

$$G_J(D_P) = \operatorname{im} G_J \cap U_P = W_P,$$

so the restricted map is onto W_P. □

The first isomorphism theorem applied to the restricted map gives $D_P/K_P \cong W_P$.

Proposition 3.2 (Canonical puncturing–shortening description)**.** *With puncturing and shortening taken onto the helper set J,*

$$D_P = \mathrm{punct}_J(I^\perp), \qquad K_P = \mathrm{short}_J(I^\perp). \tag{2}$$

Thus punct_J restricts to J, while short_J first requires the coordinates on P to vanish and then restricts to J. These operations are dual to one another. Taking duals inside $\mathbb{F}_q^J$ gives

$$D_P^\perp = \mathrm{short}_J(I), \qquad K_P^\perp = \mathrm{punct}_J(I). \tag{3}$$

Consequently the nested pair depends only on the code I and the split $E = P \sqcup J$; changing the row basis of a generator matrix leaves it unchanged.

Proof. A vector $a \in \mathbb{F}_q^J$ lies in $\mathrm{punct}_J(I^\perp)$ exactly when some $p \in \mathbb{F}_q^P$ satisfies $G_P p + G_J a = 0$, equivalently $G_J a \in \mathrm{im}\, G_P$, which is $a \in D_P$. It lies in $\mathrm{short}_J(I^\perp)$ exactly when $(0, a) \in I^\perp$, equivalently $G_J a = 0$, which is $a \in K_P$. The dual identities follow from the standard puncturing–shortening duality. ☐

Thus a monomial relabeling of helpers gives the corresponding monomially equivalent pair, while a row-basis change does nothing to the pair itself.

For a subspace $A \leq \mathbb{F}_q^J$, write $\mathrm{supp}\, A$ for the union of the supports of its vectors. For nested linear codes $K \subseteq D \leq \mathbb{F}_q^J$, the tth relative generalized Hamming weight is

$$M_t(D, K) = \min\{|\mathrm{supp}\, L| : L \leq D, \dim L - \dim(L \cap K) = t\}.$$

Equivalently, one may minimize over subspaces disjoint from K and of dimension t. Strict growth and the relative Singleton bound are Proposition 2 and Theorem 4 of Luo et al. [26]; standard treatments also cover ordinary Wei duality and computational forms [14, 37]. For nested code pairs in linear secret sharing, RDLPs and RGHWs already give exact information and reconstruction thresholds [22].

The following result is the standard nested-code support invariant applied to the canonical puncturing–shortening pair, with the normalization and helper support correspondence stated in recovery language.

Theorem 3.3 (Relative weights are exact recovery costs)**.** *Let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The minimum helper-union size among recovery systems for all helper-recoverable t-dimensional subspaces of target linear combinations, denoted $\mu_t(P)$, is*

$$\mu_t(P) = M_t(D_P, K_P).$$

More precisely, normalized recovery systems of recovered dimension t are in support-preserving correspondence, after forgetting the target normalization, with subspaces

$$L \leq D_P, \qquad \dim L = t, \qquad L \cap K_P = 0,$$

while the target normalization is a linear section through G_P. More precisely, it is a section of G_P over the chosen recovered subspace $T \leq W_P$.

Proof. Let a normalized system recover a t-dimensional subspace $T \le W_P$. In the notation of Section 2, put $L = \beta(T)$. Since $G_J\beta = -\operatorname{id}_T$, the map β is injective, $L \le D_P$, and $L \cap K_P = 0$. Moreover $G_J L = T$, and the union of helpers in the system is exactly $\operatorname{supp} L$.

Conversely, let $L \le D_P$ have dimension t and meet K_P trivially. Then $G_J|_L$ is an isomorphism from L onto the t-dimensional subspace $T = G_J L \le W_P$. Choose a linear section $\alpha : T \to \mathbb{F}_q^P$ with $G_P\alpha = \operatorname{id}_T$. Put $\beta = -(G_J|_L)^{-1}$. Then $G_P\alpha + G_J\beta = 0$, so (α, β) is a normalized recovery system with helper union $\operatorname{supp} L$.

It remains to compare this minimum with the standard relative-weight definition. If $L' \le D_P$ satisfies

$$\dim L' - \dim(L' \cap K_P) = t,$$

choose a vector-space complement L to $L' \cap K_P$ inside L'. Then $\dim L = t$, $L \cap K_P = 0$, and $\operatorname{supp} L \subseteq \operatorname{supp} L'$. The reverse inclusion of feasible classes is immediate. The two minima are equal. $\square$

The corresponding relative dimension/length profile is

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H| = s}} \big(\dim(D_P \cap \mathbb{F}_q^H) - \dim(K_P \cap \mathbb{F}_q^H)\big),$$

where $\mathbb{F}_q^H$ denotes vectors supported on H.

Proposition 3.4 (Recovery interpretation of the relative profile)**.** *For $0 \le s \le |J|$,*

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H| = s}} \dim\big(W_P \cap \operatorname{span}(G_H)\big).$$

Consequently,

$$M_t(D_P, K_P) = \min\{s : K_s(D_P, K_P) \ge t\}.$$

Thus K_s is the maximum number of independent target linear combinations recoverable from a fixed budget of s helpers.

Proof. Fix H. Restrict the exact map (1) to $D_P \cap \mathbb{F}_q^H$. Its kernel is $K_P \cap \mathbb{F}_q^H$, and its image is $W_P \cap \operatorname{span}(G_H)$. Rank–nullity gives the first equality after maximizing over H. The second is the standard inverse relation between the relative dimension/length profile and relative generalized weights; it also follows directly from Theorem 3.3. $\square$

The shortening–puncturing duality behind the relative profiles of nested codes is standard in wiretap and ramp-secret-sharing theory [26, Lemma 1 and Theorem 2]; see also [14, eqs. (1)–(2)]. Its recovery interpretation gives the complementary question to Theorem 3.3: instead of minimizing helpers needed to recover t dimensions, it minimizes failures that leave t dimensions ambiguous.

Proposition 3.5 (Failures and residual ambiguity)**.** *Let $\ell = \dim W_P$. By (3), the dual nested pair is $\operatorname{short}_J(I) \subseteq \operatorname{punct}_J(I)$. If the helpers in $H \subseteq J$ survive and those in $F = J \setminus H$ fail, then the unrecovered target dimension is*

$$\ell - \dim\big(W_P \cap \operatorname{span}(G_H)\big) = \dim(K_P^\perp \cap \mathbb{F}_q^F) - \dim(D_P^\perp \cap \mathbb{F}_q^F). \tag{4}$$

Consequently, for $1 \le t \le \ell$,

$$M_t(K_P^\perp, D_P^\perp) = \min\left\{|F| : \dim\big(W_P \cap \operatorname{span}(G_{J\setminus F})\big) \le \ell - t\right\}. \tag{5}$$

Thus the dual relative weight is the minimum number of helper failures that leave at least t target dimensions ambiguous.

Proof. For a code $C \leq \mathbb{F}_q^J$, puncturing onto F gives

$$\dim C - \dim(C \cap \mathbb{F}_q^H) = \dim(C|_F) = |F| - \dim(C^\perp \cap \mathbb{F}_q^F).$$

The first difference counts exactly the dimensions lost when coordinates in H are deleted, namely the quotient by the subcode supported on H. Apply this identity to D_P and K_P and subtract. Since $\dim D_P - \dim K_P = \ell$, Proposition 3.4 gives (4). The right-hand side is the relative dimension of the dual pair $D_P^\perp \subseteq K_P^\perp$ on F. Its least support size at value at least t is $M_t(K_P^\perp, D_P^\perp)$: a subspace attaining the relative weight is supported on such an F, while any relative dimension at least t admits a t-dimensional subspace disjoint from $D_P^\perp \cap \mathbb{F}_q^F$. This proves (5). $\square$

The strict inequalities

$$1 \leq M_1(D_P, K_P) < \cdots < M_\ell(D_P, K_P)$$

therefore say that each additional independent target combination requires a strictly larger helper union. These minima solve the single-block problem but do not yet provide recursive state: the next section restores the functional labels needed for exact concatenation. Only when an outer-distance condition removes every nonzero label does the answer collapse to $M_t(D_P, K_P) + d(I^\perp)$.

4 Exact confinement under concatenation

The argument has two stages. First, every concatenated recovery system is classified by its outer functional label. The zero label gives a local recovery system plus an inner-dual perturbation; each nonzero label prescribes a coset in every active block. Minimizing these two sectors gives the exact finite nonconfinement cost. Second, the full label-indexed cost functions, rather than their global scalar minima, are passed to the next concatenation level by an associative min–sum law.

Represented inner code. Let $L/\mathbb{F}_q$ be a finite extension and let $\iota : L \xrightarrow{\sim} I \leq \mathbb{F}_q^E$ be an $\mathbb{F}_q$-linear isomorphism, used as the inner encoder. Assume $0 < \dim I < |E|$. Whenever the outer alphabet L occurs below, I denotes the represented inner code (I, ι), not only its image subspace. In particular, $O \circ I$, Φ_I, $\lambda_{I,T}$, $\mu_{I,P,T}$, and $\Gamma_{j,T}(O, I)$ depend on ι. The earlier pair $K_P \subseteq D_P$ depends only on the image code and target/helper split. Proposition 4.2 shows that this distinction cannot be suppressed under concatenation.

For an L-linear outer code $O \leq L^N$, write

$$O \circ I = \{(\iota(a_1), \ldots, \iota(a_N)) : (a_1, \ldots, a_N) \in O\}.$$

Each block has a label map

$$\underbrace{\mathbb{F}_q^E}_{\text{inner coefficients}} \xrightarrow{\Phi_I} \underbrace{L^*}_{\text{functional label}}, \qquad \Phi_I(y)(a) = \langle y, \iota(a) \rangle.$$

Here $L^* = \mathrm{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q)$. Two coefficient vectors receive the same label precisely when they differ by an inner dual word, because $\ker \Phi_I = I^\perp$. The prescribed-coset costs below record the cheapest coefficient support in each fibre of this map.

Outer compatibility. A block vector $y = (y_1, \dots, y_N)$ belongs to $(O \circ I)^{\perp}$ precisely when

$$(\Phi_I(y_1), \dots, \Phi_I(y_N)) \in \mathrm{FD}(O), \tag{6}$$

where

$$\mathrm{FD}(O) = \{(\beta_j) \in (L^*)^N : \textstyle\sum_j \beta_j(a_j) = 0 \text{ for every } (a_j) \in O\}.$$

Indeed, both conditions say that y is orthogonal to every encoded outer word. Write $\beta_j(a) = \mathrm{Tr}_{L/\mathbb{F}_q}(b_j a)$. The functional-dual condition becomes

$$\mathrm{Tr}_{L/\mathbb{F}_q}\left(\sum_j b_j a_j\right) = 0 \qquad \text{for every } (a_j) \in O.$$

Because O is L-linear, the same holds after multiplying (a_j) by every $\lambda \in L$. Nondegeneracy of the trace pairing then forces $\sum_j b_j a_j = 0$. Hence $(b_j) \in O^{\perp}$, and the converse is immediate. This identification preserves block support, so the support distance of $\mathrm{FD}(O)$ is $d(O^{\perp})$.

Confinement and target normalization. Fix a target block and a nonzero target-message subspace $T \le U_P$. A system is *confined* if every one of its equations is supported in that target block. Let $\rho_T(I)$ be the minimum helper-union size of a normalized recovery system for T in the inner code, with value ∞ when no such system exists. Thus $\rho_T(I) < \infty$ exactly when $T \le W_P$. In a concatenation, a normalized system for T means a linear family of dual equations whose target-block coefficient map $\alpha : T \to \mathbb{F}_q^P$ satisfies $G_P \alpha = \mathrm{id}_T$ under the fixed inner-message identification. This agrees with the intrinsic target image of the concatenated generator whenever the outer projection onto the chosen block is surjective. In particular, $d(O^{\perp}) > 1$ guarantees surjectivity: an L-linear coordinate image is either 0 or L, and a zero image would put a weight-one word in $O^{\perp}$.

For an $\mathbb{F}_q$-linear map $B : T \to L^*$, define

$$\lambda_{I,T}(B) = \min_{\substack{Y : T \to \mathbb{F}_q^E \text{ linear} \\ \Phi_I Y = B}} |\operatorname{supp} Y(T)|. \tag{7}$$

For the target block, define

$$\mu_{I,P,T}(B) = \min_{\substack{\alpha : T \to \mathbb{F}_q^P,\ \beta : T \to \mathbb{F}_q^J \text{ linear} \\ G_P \alpha = \mathrm{id}_T \\ \Phi_I(\alpha, \beta) = B}} |\operatorname{supp} \beta(T)|. \tag{8}$$

An empty minimum is ∞. After choosing a basis of T, (7) is the minimum union support of representatives of prescribed cosets of $I^{\perp}$. This is the fixed-instance quantity underlying generalized coset weights and generalized covering radii [42, 10]. The target version retains the normalization required by the recovery problem. In particular,

$$\mu_{I,P,T}(0) = \rho_T(I). \tag{9}$$

For each equation parameter $t \in T$, the value $B(t)$ is exactly the outer functional label induced by that equation in the block.

Let

$$\mathcal{B}_T(O) = \mathrm{Hom}_{\mathbb{F}_q}\big(T, \mathrm{FD}(O)\big).$$

For $B \in \mathcal{B}_T(O)$, write $B = (B_h)_{h=1}^N$ with $B_h : T \to L^*$. Assume $N \geq 2$, fix a target block j whose coordinate projection $O \to L$ is nonzero, hence surjective, and put

$$\Gamma_{j,T}(O, I) = \min\left\{ \mu_{I,P,T}(0) + d(I^\perp), \min_{0 \neq B \in \mathcal{B}_T(O)} \left(\mu_{I,P,T}(B_j) + \sum_{h \neq j} \lambda_{I,T}(B_h) \right) \right\}. \tag{10}$$

Theorem 4.1 (Exact nonconfinement cost from labelled coset supports)**.** *Under the hypotheses above, $\Gamma_{j,T}(O, I)$ is the least helper-union size of a nonconfined normalized recovery system for T in $O \circ I$, with value ∞ if no such system exists. Thus every cost-$\leq r$ system for T is confined if and only if*

$$r < \Gamma_{j,T}(O, I). \tag{11}$$

For $1 \leq t \leq \ell = \dim W_P$, set

$$\Gamma_{j,t}(O, I) = \min_{\substack{T \leq W_P \\ \dim T = t}} \Gamma_{j,T}(O, I). \tag{12}$$

Then every cost-$\leq r$ system for every helper-recoverable t-dimensional target subspace is confined if and only if $r < \Gamma_{j,t}(O, I)$. Below either threshold, restriction and zero-extension are inverse bijections on normalized systems and preserve exact helper supports.

Proof. Write a recovery system as an $\mathbb{F}_q$-linear map $Y : T \to (O \circ I)^\perp$ and let Y_h be its block maps. By (6),

$$B_h = \Phi_I Y_h$$

defines a linear map $B : T \to \mathrm{FD}(O)$. Conversely, compatible linear lifts of the block maps of any such B give a map into the concatenated dual. At the target block, compatibility with the chosen target space is exactly the constraint in (8).

Suppose first that $B \neq 0$. No nonzero element of $\mathrm{FD}(O)$ is supported only at j: testing such an element against outer words whose jth coordinate ranges over L would force its jth functional to vanish. Hence some block map B_h with $h \neq j$ is nonzero, and every lift is nonconfined. Distinct inner blocks have disjoint helper coordinates, so the minimum helper union for this fixed B is

$$\mu_{I,P,T}(B_j) + \sum_{h \neq j} \lambda_{I,T}(B_h).$$

The block lifts can be chosen independently, so this lower bound is attained.

If $B = 0$, every block map takes values in $I^\perp = \ker \Phi_I$. The target block costs $\rho_T(I) = \mu_{I,P,T}(0)$. Nonconfinement requires a nonzero map $T \to I^\perp$ in another block. Its image has union support at least $d(I^\perp)$, and equality is attained by $u \mapsto f(u)v$, where $0 \neq f \in T^*$ and v is a minimum-weight word of $I^\perp$.

The zero and nonzero functional sectors are exhaustive, proving (10) and (11). Minimizing over t-dimensional T proves the ranked statement. Below the first nonconfined cost, every global system has zero coefficient maps in all blocks outside j. Restriction therefore preserves its target normalization and helper support. Conversely, zero-extending an inner dual system remains orthogonal to every concatenated word, so the two operations are inverse with coefficients and supports unchanged. $\square$

A complete small instance. Let $I \le \mathbb{F}_2^3$ have generator columns $G = (e_1 \mid e_2, e_1 + e_2)$, with the first coordinate targeted. Then $K_P = 0$, $D_P = \langle(1,1)\rangle$, and $M_1(D_P, K_P) = 2$: the target is recovered from exactly the two helpers. Moreover $I^\perp = \langle(1,1,1)\rangle$, so the zero-functional nonconfinement cost is $2 + 3 = 5$.

Identify the two-dimensional message space with $L = \mathbb{F}_4$ and concatenate with the length-six single-parity-check code over L. Its functional dual is the repetition line. Every nonzero label is active in all five blocks outside the target block. An active block has a nonzero functional label, so every lift has nonempty support and costs at least one helper. Thus its nonzero-functional sector also costs at least five, so $\Gamma_{j,1} = 5$. Every recovery system of cost at most four is therefore confined, while a cost-five witness is obtained by adjoining the inner-dual word $(1,1,1)$ in another block.

The later scheduler uses the same support data but asks a different question. Two demands in distinct blocks fit unit helper capacities, whereas two demands in one block compete for the same helper pair. Thus this instance will serve as the model case for both composition and scheduling without identifying the two optimizations.

4.1 Repeated concatenation

The ordinary prescribed-coset costs retain the intermediate labels needed at the next concatenation level. Target-normalized composition also retains the part of each intermediate label already supplied by target coordinates. In schematic form,

$$\text{cost of output label} = \min_{\text{compatible intermediate labels}} \sum_{\text{inner blocks}} \text{inner realization cost.}$$

The eliminated variables are the intermediate labels shared by adjacent levels. The resulting formulas are instances of variable elimination over the min-sum semiring [2]. This is also the algebraic form of passing symbol-indexed inner costs to an outer decoder, as in soft and generalized-concatenated decoding [17, 5].

The zero-functional nonconfinement cost alone is not compositional. The next separation holds the underlying binary code and all its unlabelled recovery data fixed.

Proposition 4.2 (Functional labels are necessary for finite composition)**.** *Over $\mathbb{F}_2 \subseteq \mathbb{F}_4$, there are two represented inner encoders with the same image code, dual code, target/helper nested pair, complete relative-weight hierarchy, dual distance, unlabelled normalized helper-support family, and zero-functional nonconfinement costs for every recoverable target subspace, but one fixed outer represented code gives different exact nonconfinement costs.*

Proof. Write the nonzero elements of $\mathbb{F}_4$ as $1, \omega, \omega^2 = 1 + \omega$ and take generator columns

$$I_1 : (1, 1, \omega), \qquad I_2 : (1, 1, \omega^2),$$

with the first coordinate targeted. Relative to the binary basis $(1, \omega)$ of $\mathbb{F}_4$, each displayed field element is a two-dimensional binary generator column. Both generator matrices have binary row space

$$\{(x, x, y) : x, y \in \mathbb{F}_2\}$$

and dual $\langle(1,1,0)\rangle$. Thus both have dual distance 2, $K_P = 0 \subseteq D_P = \langle(1,0)\rangle$, $M_1 = 1$, and the same unique minimal helper support. Their recoverable target space is one-dimensional, so their complete zero-functional profile is the common value $1 + 2 = 3$.

In the label order $(1, \omega, \omega^2)$, the ordinary coset costs are $(1,1,2)$ for I_1 and $(1,2,1)$ for I_2; the normalized target costs are respectively $(0,2,1)$ and $(0,1,2)$. Take the fixed outer code whose functional dual is the line spanned by $(1, \omega)$. Its nonzero multiples have target label s and external label $s\omega$. The complete calculation is

s	s	$s\omega$	$\mu_1(s)$	$\lambda_1(s\omega)$	total	$\mu_2(s)$	$\lambda_2(s\omega)$	total
1	1	ω	0	1	1	0	2	2
ω	ω	ω^2	2	2	4	1	1	2
ω^2	ω^2	1	1	1	2	2	1	3

Equation (10) therefore gives exact costs 1 and 2. Only the alignment of costs with the intermediate functional labels has changed. □

Let $\mathbb{F}_q \subseteq L \subseteq M$ be a tower of finite fields. Let B be an $\mathbb{F}_q$-linear represented code with message space L, and let A be an L-linear represented code with message space M. The composite $A \circ B$ replaces each coordinate of an A-word by its represented B-block. After identifying a functional with its unique trace coefficient, write their dual restriction maps as

$$\phi_B : \mathbb{F}_q^{E_B} \longrightarrow L, \qquad \phi_A : L^{E_A} \longrightarrow M.$$

For an $\mathbb{F}_q$-space T and $b : T \to L$, put

$$\Lambda_{B,T}(b) = \min_{\phi_B Y = b} |\operatorname{supp} Y(T)|.$$

Define $\Lambda_{A,T}$ and $\Lambda_{A\circ B,T}$ similarly. If $J \subseteq E_B$, let $\Lambda_{B,J,T}$ denote the same minimum with Y supported on J and ϕ_B restricted to those coordinates. Thus ϕ is the trace-coordinate form of Φ, and Λ is the corresponding ordinary prescribed-coset cost λ. The change of letters only distinguishes the two levels in the following substitution.

Theorem 4.3 (Min–sum closure of prescribed-coset support costs). *For every $\mathbb{F}_q$-linear map $c : T \to M$,*

$$\Lambda_{A\circ B,T}(c) = \min_{X:T\to L^{E_A}} \min_{\phi_A X = c} \sum_{e\in E_A} \Lambda_{B,T}(X_e). \tag{13}$$

The same formula iterates associatively: either parenthesization of a finite tower minimizes over the same intermediate labels and sums the same leaf costs.

Proof. A leaf-level lift $Y : T \to \mathbb{F}_q^{E_A\times E_B}$ determines intermediate maps $X_e = \phi_B Y_e$. Its support is the disjoint union of its supports in the inner blocks. At fixed X, the block lifts are independent and attain their minima separately, which proves (13). At three or more levels, either parenthesization eliminates the same intermediate arrays, proving associativity. □

Sharp scalar envelopes. The full labelled function gives simple bounds when only its smallest and largest nonzero values are retained. When $T \neq 0$, set

$$\delta_{B,T} = \min_{0\neq b:T\to L} \Lambda_{B,T}(b), \qquad R_{B,T} = \max_{0\neq b:T\to L} \Lambda_{B,T}(b).$$

For $c \neq 0$, the exact formula has the sharp coarsening

$$\delta_{B,T}\Lambda_{A,T}(c) \le \Lambda_{A\circ B,T}(c) \le R_{B,T}\Lambda_{A,T}(c). \tag{14}$$

Neither constant can be improved without further hypotheses. Indeed, every nonzero intermediate coordinate map costs between $\delta_{B,T}$ and $R_{B,T}$. Minimizing the number of nonzero intermediate coordinates proves the bounds; a one-coordinate outer realization attaining either extremum proves sharpness.

Target-normalized closure. For recovery, ordinary labels are supplemented by the contribution already supplied on the target coordinates. Suppose that a target set in $A \circ B$ meets block e in $P_e \subseteq E_B$, and put $J_e = E_B \setminus P_e$ and $U_{P_e} = \operatorname{im} \phi_{B,P_e}$. If $T \leq M$ and $\iota_T : T \hookrightarrow M$ is the inclusion, the variables in the next formula have the following roles:

$$\begin{array}{r|l} U_e & \text{intermediate contribution supplied by target coordinates in block } e \\ X_e & \text{total intermediate functional label required in block } e \\ X_e - U_e & \text{label that the helper coordinates in block } e \text{ must realize.} \end{array}$$

The target-normalized cost is therefore

$$\mu_{A\circ B,P,T}(c) = \min_{\substack{U,X:T\to L^{E_A} \\ U_e(T)\subseteq U_{P_e}\ (e\in E_A) \\ \phi_A U=\iota_T,\ \phi_A X=c}} \sum_{e\in E_A} \Lambda_{B,J_e,T}(X_e - U_e). \tag{15}$$

To verify the formula, lift U_e to an actual target map $a_e : T \to \mathbb{F}_q^{P_e}$. The condition $\phi_A U = \iota_T$ is exactly target normalization. Once X_e is fixed, the helper contribution must lift $X_e - U_e$ through ϕ_{B,J_e}, independently in each block. If one also stores the target maps a_e and minimizing helper lifts, backtracking recovers one coefficient-level witness. Retaining every witness requires the full lift sets, not only these numerical minima.

Dual-distance recursion. The same zero/nonzero-label split gives

$$d((A\circ B)^\perp) = \min\left\{ d(B^\perp),\ \min_{0\neq z\in A^\perp} \sum_{e\in E_A} \Lambda_{B,\mathbb{F}_q}(u\mapsto uz_e) \right\}. \tag{16}$$

A word in $(A \circ B)^\perp$ either has zero intermediate label, when a minimum word of $B^\perp$ in one block is optimal, or has a nonzero label $z \in A^\perp$, when the block representatives minimize independently. This proves (16).

Tower consequence. Substituting these recursively computed costs into (10) gives the exact nonconfinement cost through any finite tower, with the zero and nonzero functional sectors kept separate at every level.

The trace identifications in these formulas are compatible with the tower. Write $\iota_A : M \to A \leq L^{E_A}$ for the represented encoder of A. For $v \in T$ and $m \in M$, trace transitivity gives

$$\sum_e \operatorname{Tr}_{L/\mathbb{F}_q}\big(X_e(v)\,\iota_A(m)_e\big) = \operatorname{Tr}_{M/\mathbb{F}_q}\big(\phi_A(X(v))m\big),$$

and nondegeneracy of the trace pairing identifies the composite restriction map with $\phi_A \circ \phi_B^{E_A}$, where $\phi_B^{E_A}$ applies ϕ_B blockwise.

4.2 Contextual equivalence and rank reduction

For rank-one targets, the exact quotient of the labelled functions observable by outer concatenation has a projective description. Fix a line $T = \langle u \rangle \leq W_P$. For $b \in L$, let $\tau_b : T \to L^*$ be defined by

$$\tau_b(au)(x) = a \operatorname{Tr}_{L/\mathbb{F}_q}(bx) \qquad (a \in \mathbb{F}_q,\ x \in L),$$

and put $z_{I,T} = \rho_T(I) + d(I^\perp)$. If $[b] = [b_1 : \cdots : b_N] \in \mathbf{P}(L^N)$ has a nonzero coordinate outside the target block j, its projective class records an outer functional-dual line. Multiplying by $s \in L^\times$ changes the chosen generator of that line but not the outer context, so the probe minimizes over this scalar choice:

$$\Theta_{I,T}([b]) = \min_{s \in L^\times} \left(\mu_{I,P,T}(\tau_{sb_j}) + \sum_{h \neq j} \lambda_{I,T}(\tau_{sb_h}) \right). \tag{17}$$

Changing u or the projective representative of $[b]$ does not change this value.

Here an *outer context for the rank-one target T* is a triple (N, j, O) with $N \geq 2$, $O \leq L^N$ an L-linear code, and nonzero projection onto the distinguished block j. The context family ranges over every such N and j, while the inner target/helper split and the identified target line T remain fixed. Thus "rank one" refers to $\dim_{\mathbb{F}_q} T = 1$, not to the dimension of O. Two represented inner codes are *contextually equivalent at T* when their values of $\Gamma_{j,T}$ agree in every member of this family. A compatible further concatenation adds outer layers above the same distinguished leaf and does not retarget that leaf.

Theorem 4.4 (Minimal contextual quotient at rank one)**.** *Let $D = O^\perp \leq L^N$ under the trace identification, and suppose the projection of O onto block j is nonzero. Then*

$$\Gamma_{j,T}(O, I) = \min \left\{ z_{I,T}, \min_{\substack{Lb \leq D \\ b \neq 0}} \Theta_{I,T}([b]) \right\}, \tag{18}$$

where Lb is the one-dimensional L-subspace spanned by b, and the inner minimum is empty when $D = 0$.

Consequently, represented inner codes over the same extension field with identified target lines are contextually equivalent at T if and only if their values $z_{I,T}$ agree and the family of truncated profiles

$$[b] \longmapsto \min\{z_{I,T}, \Theta_{I,T}([b])\}$$

agrees for every $N \geq 2$, target block j, and projective tuple with nonzero external part. Thus equality of this response family realizes the exact numerical contextual quotient at rank one: no coarser numerical summary can predict the costs in every outer context. For each fixed outer length N the test family is finite; contextual equivalence quantifies over these finite families for all N rather than claiming one arity-independent finite table. It is enough to test the full outer code and outer codes whose functional dual has L-dimension one. Moreover, contextual equivalence at T is a congruence under the compatible further concatenations defined above.

Proof. A nonzero map from the one-dimensional $\mathbb{F}_q$-space T into D is determined by one nonzero tuple $b \in D$. Partition $D \setminus \{0\}$ into its L-lines in the nonzero-functional sector of (10). Minimization on the line Lb is exactly (17), which proves (18).

It remains to show that every stated probe is an actual outer-code context. The full outer code $O = L^N$ has $D = 0$ and exposes $z_{I,T}$. Given a projective tuple $[b]$ with nonzero external part, take $D = Lb$ and $O = D^\perp$. Its target projection is nonzero: otherwise D would contain the target coordinate line, forcing every external coordinate of b to vanish. This context exposes $\min\{z_{I,T}, \Theta_{I,T}([b])\}$. Equality in every context therefore forces equality of the displayed response family; the first part of the theorem gives the converse.

For the congruence statement, place the same distinguished target leaf in two composites $A \circ I_1$ and $A \circ I_2$. Here compatible means that later layers keep this same target leaf and its induced target image. Any such later context C tests the target-normalized composites $C \circ (A \circ I_i) =$

$(C \circ A) \circ I_i$. The target-normalized composition formula (15) carries the same leaf target image on both parenthesizations, and $C \circ A$ is one of the contexts quantified above. Contextual equivalence of I_1 and I_2 therefore gives equal costs after every such C. Hence contextual equivalence is a congruence for the specified compatible concatenations. □

This theorem compares numerical costs only. Full coefficient witnesses still require the minimizing lifts described after Theorem 4.3.

When $\dim_{\mathbb{F}_q} T = t$, the image of any map $B : T \to \mathrm{FD}(O)$ lies in an L-subspace generated by at most t images. This observation bounds the dimension of the outer contexts needed at every target rank.

Proposition 4.5 (Rank-bounded outer tests)**.** *Let $\dim_{\mathbb{F}_q} T = t$, put $D = O^{\perp}$, and retain the nonzero projection onto block j. For L-subspaces $D' \leq D$, set $O_{D'} = (D')^{\perp} \leq L^N$. Then*

$$\Gamma_{j,T}(O, I) = \min_{\substack{D' \leq D \\ \dim_L D' \leq t}} \Gamma_{j,T}(O_{D'}, I). \tag{19}$$

Thus outer codes with functional-dual dimension at most t form a complete test family for exact contextual equivalence at target rank t.

Proof. For $B : T \to D$, let $D_B = \mathrm{span}_L B(T)$. The image of an $\mathbb{F}_q$-basis of T generates D_B over L, so $\dim_L D_B \leq t$. Regarding B as a map into D_B changes none of its block labels or costs. Conversely, every map into $D' \leq D$ is a map into D. Minimizing the same cost functional in both directions proves (19); $D' = 0$ supplies the zero-functional sector. No $D' \leq D$ contains a nonzero target-supported vector, so every $O_{D'}$ retains nonzero target projection. □

The simultaneous confinement criterion across all recoverable ranks has a single bottleneck.

Corollary 4.6 (Rank-one criterion for complete bounded transfer)**.** *For every target block j with nonzero outer projection,*

$$\min_{0 \neq T \leq W_P} \Gamma_{j,T}(O, I) = \Gamma_{j,1}(O, I). \tag{20}$$

For every integer $r \geq 0$, the following are equivalent:

1. *every cost-$\leq r$ system on every helper-recoverable target line is confined;*
2. *every cost-$\leq r$ system on every nonzero helper-recoverable target subspace is confined; and*
3. *$r < \Gamma_{j,1}(O, I)$.*

Under these conditions, restriction and zero-extension are inverse bijections simultaneously for every nonzero $T \leq W_P$, preserving every coefficient and exact helper support through radius r.

Consequently, any statistic defined solely from these bounded systems and invariant under this bijection is preserved. In particular, the bounded reliability laws and the capacity and scheduling regions defined later from their coefficients or supports agree. Targetwise and rankwise minimum costs also agree after truncation $c \mapsto \min\{c, r+1\}$ (with $\min\{\infty, r+1\} = r+1$). If every relevant inner minimum is at most r, then the untruncated minima agree as well.

Proof. If a system on $0 \neq T \leq W_P$ is nonconfined, some external block map is nonzero on a vector $u \in T$. Restriction to $\langle u \rangle$ remains nonconfined and cannot enlarge its helper union. This proves the displayed equality, and Theorem 4.1 gives the three equivalent conditions. When they hold, every bounded system is confined, so restriction deletes only zero external blocks and zero-extension is its inverse. The coefficient, support, reliability, capacity, and scheduling assertions follow from this bijection. For a minimum helper-union cost, either the inner minimum is at most r and the bijection gives equality, or both inner and concatenated minima exceed r; this is precisely equality after truncation at $r+1$. $\square$

4.3 Outer-distance collapse and bounded transfer

The exact optimization also recovers the outer-distance collapse without choosing a radius. Indeed, every nonzero functional sector costs at least $d(O^{\perp})-1$: its label has at least $d(O^{\perp})$ active blocks, the target block may contribute zero helpers, and every other active block contributes at least one. The zero-functional sector attains $M_t(D_P, K_P) + d(I^{\perp})$ after minimization over T. Consequently,

$$d(O^{\perp}) \geq M_t(D_P, K_P) + d(I^{\perp}) + 1 \quad \Longrightarrow \quad \Gamma_{j,t}(O, I) = M_t(D_P, K_P) + d(I^{\perp}). \tag{21}$$

Theorem 4.7 (Confinement for a fixed target subspace)**.** *Let r be a nonnegative integer and let $O \leq L^N$ be an L-linear outer code with $N \geq 2$ and $d(O^{\perp}) > r+1$, and assume $T \leq W_P$. Then $O \circ I$ confines every normalized recovery system for T of cost at most r if and only if*

$$r < \rho_T(I) + d(I^{\perp}). \tag{22}$$

Below this bound, zero-extension gives a support-preserving bijection between the inner systems of cost at most r and the corresponding systems in the concatenation. Hence the same conclusions hold for all sufficiently long members of any outer family with dual distance tending to infinity, at every fixed r.

Proof. If $0 \neq B \in \mathcal{B}_T(O)$, choose $u \in T$ with $B(u) \neq 0$. The tuple $B(u) \in \mathrm{FD}(O)$ has at least $d(O^{\perp})$ nonzero blocks. At most one is the target block, so every lift has at least $d(O^{\perp}) - 1 > r$ helpers outside it. Thus no nonzero-functional sector in (10) occurs at cost at most r.

The zero-functional sector has exact cost $\mu_{I,P,T}(0) + d(I^{\perp}) = \rho_T(I) + d(I^{\perp})$. Theorem 4.1 now gives both directions of (22) and the support-preserving bijection. $\square$

Minimizing the fixed-subspace threshold over all recovered subspaces of fixed dimension gives the promised hierarchy.

Theorem 4.8 (Confinement by recovered dimension under an outer-distance condition)**.** *Let r be a nonnegative integer, let $O \leq L^N$ be L-linear with $N \geq 2$ and $d(O^{\perp}) > r+1$, and let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The concatenation $O \circ I$ confines every cost-$\leq r$ recovery system for every helper-recoverable t-dimensional target subspace if and only if*

$$r < M_t(D_P, K_P) + d(I^{\perp}). \tag{23}$$

When (23) holds, restriction and zero-extension preserve every normalized equation and its exact helper support. For an outer family with dual distance tending to infinity, these conclusions hold for all sufficiently long members at every fixed r.

Proof. By Theorem 3.3, every such target subspace has inner cost at least $M_t(D_P, K_P)$, and some target subspace attains this minimum. Apply Theorem 4.7 to each subspace. The lower bound gives sufficiency uniformly; an attaining subspace and the external rank-one perturbation in that proof give necessity. $\square$

4.4 One-coordinate specialization

For $\beta \in L^*$, put

$$\lambda_I(\beta) = \min_{\Phi_I(y)=\beta} \mathrm{wt}(y), \qquad \mu_{I,x}(\beta) = \min_{\substack{\Phi_I(y)=\beta \\ y_x \neq 0}} \mathrm{wt}(y),$$

with value ∞ when a constrained fibre is empty. The first quantity is the usual coset-leader weight for the inner encoder, while the second retains the target-coordinate constraint. The functional-dual decomposition itself is classical concatenation machinery [8, Theorem 2.1]. Here a dual word "through x" simply means that its coefficient at x is nonzero. The quantity below counts total dual-word weight; the general recovery cost excludes the forced target coordinate, so for a nonzero target column the two conventions satisfy $Z_{j,x} = \Gamma_{j,T} + 1$.

Theorem 4.9 (Exact weighted finite-length confinement at one coordinate)**.** *Fix an outer block j, an inner coordinate x, and $N \geq 2$, and assume that the coordinate projection $O \to L$ at block j is nonzero, hence surjective. The minimum total Hamming weight of a word in $(O \circ I)^\perp$ that is nonzero at (j,x) and is not the zero-extension of a word of $I^\perp$ in block j is*

$$Z_{j,x}(O,I) = \min \left\{ \mu_{I,x}(0) + d(I^\perp), \min_{\substack{\beta=(\beta_h)\in \mathrm{FD}(O) \\ \beta \neq 0}} \left(\mu_{I,x}(\beta_j) + \sum_{h\neq j} \lambda_I(\beta_h) \right) \right\}. \tag{24}$$

A minimum over an empty set is ∞. Consequently, for every $r \geq 0$, restriction to block j and zero-extension are inverse, support-preserving bijections between the normalized recovery equations through x in I and those through (j,x) in $O \circ I$, both using at most r helpers, if and only if

$$Z_{j,x}(O,I) > r + 1. \tag{25}$$

The same equivalence holds for the exact helper-support families. When these conditions hold, the inclusion-minimal recovery sets are copied as well.

Proof. Write a concatenated dual word in blocks and apply (6). For a fixed nonzero functional tuple, the block representatives minimize independently, giving the second term of (24). For the zero tuple, nonconfinement costs $\mu_{I,x}(0)$ in the target block and $d(I^\perp)$ in another block. The projection hypothesis makes confined words exactly the zero-extensions of inner-dual words, so these two sectors prove (24).

If the target column is nonzero, take $P = \{x\}$ and let $T \leq U_P$ be its span. After choosing a nonzero vector of T, a linear map $T \to L^*$ is one functional. Normalizing its target coefficient and rescaling the functional tuple identify the minima in (10) with those in (24). The target coordinate contributes one to total word weight and is not a helper, so

$$Z_{j,x}(O,I) = \Gamma_{j,T}(O,I) + 1.$$

Theorem 4.1 gives the claimed condition. If the target column is zero, the same condition follows directly from the exact minimum already proved. In either case, below the minimum every equation is confined, so restriction and zero-extension are inverse. In one dimension every exact helper support is the support of its normalized equation; the equivalence also holds for exact helper-support families and their inclusion-minimal members. □

Theorem 4.9 retains information that the outer support distance discards. The following family makes the difference strict without using a geometric inner code. Its construction chooses an outer dual repetition line whose projective multipliers avoid every ratio between low-cost inner labels. Consequently the labelled criterion holds even though ordinary distance sees no margin.

Proposition 4.10 (A family from a Singer cycle without the outer-distance condition)**.** *Let $k \geq 2$ and suppose*

$$\frac{q^k - 1}{q - 1} > (k+1)^2. \tag{26}$$

Let I be the $[k+1, k, 2]_q$ single-parity-check code generated by

$$G = (I_k \mid \mathbf{1}),$$

and identify its message space with $L = \mathbb{F}_{q^k}$. For every $k+1 \leq n \leq 2k+1$, there is an L-linear length-n outer code O with $d(O^\perp) = n$ such that (25) holds at helper radius $n-1$ for every inner coordinate and every outer block. Thus all normalized radius-$(n-1)$ recovery equations and their exact helper supports are copied blockwise although the usual outer-distance condition fails: here $d(O^\perp) = n$ is not strictly greater than $r + 1 = n$.

In particular, this holds for the $[4, 3, 2]_5$ inner code and a length-five outer code at helper radius four.

Proof. The $k+1$ coordinate functionals of the encoder give a set S of $k+1$ points in the projective space on L^*. The multiplication maps of L, modulo $\mathbb{F}_q^*$, induce on that projective space a Singer cycle of order $(q^k-1)/(q-1)$, acting regularly on its points [39]. Here regular means that a unique cycle element sends any chosen point to any other. Each ordered pair of points of S therefore forbids at most one cycle element, so at most $|S|^2 = (k+1)^2$ elements send a point of S to a point of S. Thus (26) permits a multiplication map $a : L \to L$, explicitly $a(u) = \gamma u$ for some $\gamma \in L^\times$, whose dual action satisfies $a^*(S) \cap S = \varnothing$.

Set

$$O = \{u \in L^n : u_0 + a(u_1) + u_2 + \cdots + u_{n-1} = 0\}.$$

Every functional-dual tuple has the form $(f, f \circ a, f, \ldots, f)$. If $f \neq 0$, all n entries are nonzero, so $d(O^\perp) = n$. A realization of total weight n would have weight one in every block. Weight-one realizations are exactly the scalar multiples of the $k+1$ coordinate functionals, so both $[f]$ and $[f \circ a]$ would lie in S, contrary to the choice of a. Every nonzero tuple therefore costs at least $n+1$.

Every outer coordinate projection is surjective. Finally, $I^\perp$ is spanned by $(1, \ldots, 1, -1)$, so $d(I^\perp) = \mu_{I,x}(0) = k+1$ for every x. The zero-functional cost in (24) is $2k+2 > n$, and the nonzero-functional cost is at least $n+1$. Thus $Z_{j,x}(O, I) > n$ for every j, x. At $r = n-1$ the ordinary distance condition would require $d(O^\perp) > n$, which is false, while the exact weighted condition holds. Taking $(q, k, n) = (5, 3, 5)$ gives the stated concrete instance. Since the inner recovery equations have k helpers and $n - 1 \geq k$, the asserted transfer is nonvacuous throughout the family. □

For a nondegenerate singleton target $P = \{x\}$, the first relative weight is the least number of helpers in a dual word through x. If the total-weight threshold used for one-coordinate equations is denoted by $z_x(I)$, then

$$z_x(I) = M_1(D_P, K_P) + 1 + d(I^\perp). \tag{27}$$

The helper-radius condition (23) is therefore $r + 1 < z_x(I)$, with no change of convention hidden in the reduction. Theorem 4.1 is the common finite formula. The outer-distance condition removes its nonzero-functional sector and gives the RGHW threshold in every recovered dimension. Restricting instead to one target coordinate gives Theorem 4.9 without that condition.

The labelled functions are sufficient for exact numerical composition; minimizing lifts must also be stored to propagate witnesses. Their coarsenings answer different questions. The next section shows sharply that relative-weight minima do not determine recovery reliability and that the abstract nested pair does not determine confinement.

5 Sharp limits of coarser recovery invariants

The preceding section supplies an exact numerical state for composition. We now rule out two tempting coarsenings. Relative generalized weights record minimum support sizes but not the overlap pattern of all recovery sets, which controls stochastic repair. The abstract nested pair records more than those minima, but not its ambient inner-code realization; that presentation can change $d(I^{\perp})$ and the prescribed-coset support costs in (10). The support-side loss is

$$\text{normalized recovery systems} \longrightarrow \text{exact helper supports} \longrightarrow \text{relative-weight minima.}$$

The exact finite concatenation formula instead uses the outer functional dual and the joint support costs of prescribed inner cosets. The first separation below rules out the complete RGHW hierarchy as a reliability invariant at every quotient rank. The second rules out even the full abstract pair, together with the same unlabelled helper supports, as a confinement invariant.

First consider a one-dimensional recovered space. On the helper set $\{0, \ldots, 8\}$, let

$$\mathcal{A} = \{158, 026, 045, 013, 478\},$$
$$\mathcal{B} = \{168, 237, 078, 015, 124\},$$

where, for example, 158 denotes $\{1, 5, 8\}$. Both are realizable as the minimum recovery sets at a distinguished coordinate of a represented $[10, 4, 6]_q$ code over a common finite prime field. Here a *clutter* means a family of sets none of which contains another; in this example it is the five inclusion-minimal recovery supports.

Proposition 5.1 (Rank-one support-overlap separation)**.** *The two distinguished coordinates above have the same minimum helper cost $M_1 = 3$. Under independent homogeneous helper survival with probability s, their radius-three repair reliabilities are*

$$R_{\mathcal{A}}(s) = 5s^3 - 7s^5 - s^6 + 5s^7 - s^9, \tag{28}$$
$$R_{\mathcal{B}}(s) = 5s^3 - 7s^5 - 2s^6 + 8s^7 - 3s^8. \tag{29}$$

In particular, the complete rank-one relative-weight hierarchy does not determine bounded repair reliability.

Proof. The five triples in each clutter are the minimum recovery sets, so both minimum costs are three. For $k = 1, \ldots, 5$, the numbers of k-edge subfamilies according to union size are

k	$\mathcal{A}$	$\mathcal{B}$
1	$5u^3$	$5u^3$
2	$7u^5 + 3u^6$	$7u^5 + 3u^6$
3	$2u^6 + 6u^7 + 2u^8$	$u^6 + 8u^7 + u^8$
4	$u^7 + 2u^8 + 2u^9$	$4u^8 + u^9$
5	u^9	u^9.

Inclusion–exclusion for the five edge-containment events gives (28) and (29).

We now give the representation argument. Its purpose is to realize exactly the five listed triples as collinear triples; after lifting, adjoining the distinguished target x turns exactly those triples into four-column circuits.

The two plane arrangements. Over an infinite field K, choose five projective lines $\ell_1, \dots, \ell_5$. For $\mathcal{A}$, make ℓ_2, ℓ_3, ℓ_4 concurrent and label

$$0 = \ell_2 \cap \ell_3 \cap \ell_4, \quad 1 = \ell_1 \cap \ell_4, \quad 4 = \ell_3 \cap \ell_5, \quad 5 = \ell_1 \cap \ell_3, \quad 8 = \ell_1 \cap \ell_5.$$

Choose $2, 6$ on ℓ_2, 3 on ℓ_4, and 7 on ℓ_5. For $\mathcal{B}$, make ℓ_1, ℓ_4, ℓ_5 concurrent at 1, put

$$8 = \ell_1 \cap \ell_3, \quad 7 = \ell_2 \cap \ell_3, \quad 0 = \ell_3 \cap \ell_4, \quad 2 = \ell_2 \cap \ell_5,$$

and choose $6, 3, 5, 4$ on $\ell_1, \ell_2, \ell_4, \ell_5$, respectively. The incidence skeleton is

	three concurrent lines	other prescribed intersections
$\mathcal{A}$	ℓ_2, ℓ_3, ℓ_4 at 0	$1 = \ell_1 \cap \ell_4,\ 4 = \ell_3 \cap \ell_5,\ 5 = \ell_1 \cap \ell_3,\ 8 = \ell_1 \cap \ell_5$
$\mathcal{B}$	ℓ_1, ℓ_4, ℓ_5 at 1	$8 = \ell_1 \cap \ell_3,\ 7 = \ell_2 \cap \ell_3,\ 0 = \ell_3 \cap \ell_4,\ 2 = \ell_2 \cap \ell_5$

At each free choice, avoid the finitely many unwanted joining lines. The unwanted lines are precisely those that would create an additional collinear triple. The displayed triples are therefore exactly the collinear triples.

Generic lift. Choose representatives $p_i \in K^3$ of either plane configuration, let $x = (1, 0, 0, 0)$, and lift helper i to $h_i = (t_i, p_i) \in K^4$. Dependence of the lifts over any displayed line and dependence of any four helper lifts are proper linear conditions on the t_i. For a displayed line, its three quotient vectors have a unique dependence up to scalar, and the same coefficients give one nonzero linear equation in the t_i. For four helper points, the quotient vectors span K^3, so their lifted determinant is a nonzero linear form in the four first coordinates. Avoiding this finite union makes every helper triple and quadruple independent, while the dependent four-sets through x are exactly $\{x\} \cup A$ for the five prescribed triples A.

Finite-field realization and parameters. Choose the construction over $\mathbb{Q}$ and reduce both configurations modulo one prime avoiding the finitely many nonzero determinants. The two row-span codes then have rank four, dual distance four, and minimum recovery sets exactly $\mathcal{A}$ and $\mathcal{B}$. A hyperplane not containing x contains at most three helpers because every four helpers are independent. A hyperplane containing x contains at most one displayed collinear triple, and such a triple gives four columns in a hyperplane. Thus the maximum hyperplane intersection has size four, so both codes have distance $10 - 4 = 6$. $\square$

The rank-one separation extends to every quotient rank without changing any relative generalized weight. For a nested pair $K \subseteq D \leq \mathbb{F}_q^J$ with $\dim(D/K) = \ell$, define its radius-R full-quotient reliability by

$$\Pr\{\exists L \leq D : \ L + K = D,\ |\operatorname{supp} L| \leq R,\ \operatorname{supp} L \subseteq S\},$$

where $S \subseteq J$ is the random set of independently surviving helpers, each present with probability s. The condition $L + K = D$ says that the surviving helper equations recover the entire quotient D/K; the radius R bounds their union support $|\operatorname{supp} L|$.

Theorem 5.2 (The complete RGHW hierarchy does not determine reliability)**.** *For every $\ell \geq 1$, there are two nested code pairs of quotient dimension ℓ with identical relative generalized Hamming weights $M_1, \dots, M_\ell$ but different bounded reliability laws for recovery of the full quotient.*

Proof. For $\ell = 1$, use Proposition 5.1. For $\ell > 1$, let $K_0 \subset D_0$ be either of its two associated rank-one pairs. On disjoint helper blocks, add the same $\ell - 1$ pairs

$$0 \subset \langle \mathbf{1}_{L_i} \rangle \qquad (1 \le i \le \ell - 1),$$

where each padding line has the unique support of size L_i. Thus

$$K = K_0 \oplus 0 \oplus \cdots \oplus 0, \qquad D = D_0 \oplus \langle \mathbf{1}_{L_1} \rangle \oplus \cdots \oplus \langle \mathbf{1}_{L_{\ell-1}} \rangle.$$

For a direct sum of rank-one quotient pairs, the tth relative weight is the sum of the t smallest component costs. Both constructions therefore have the same hierarchy, obtained from the common multiset $\{3, L_1, \dots, L_{\ell-1}\}$.

To verify the direct-sum formula, project a feasible helper subspace to each quotient component. A t-dimensional quotient image has nonzero projection to at least t of the one-dimensional components. On disjoint helper blocks, each active projection pays at least that component's minimum cost, so the total support is at least the sum of the t smallest costs. The direct sum of minimum words from the t cheapest components attains the bound.

Set $R = 3 + \sum_i L_i$. A recovery subspace of full quotient rank projects nontrivially to every padding component and hence uses its entire forced support. Removing those supports leaves a radius-three recovery set in the base component. Conversely, any base radius-three recovery set together with the padding generators recovers the full quotient at radius R. Thus

$$R_{\mathrm{full},R}(s) = s^{\sum_i L_i} R_{\mathrm{base},3}(s).$$

The two polynomials remain different by (28)–(29).

Every pair in the construction is realized by an inner code. Put $V = \mathbb{F}_q^J / K$, let $\pi : \mathbb{F}_q^J \to V$ be the quotient map, and choose an isomorphism $\alpha : \mathbb{F}_q^\ell \to D/K \le V$. The generator map

$$G : \mathbb{F}_q^\ell \oplus \mathbb{F}_q^J \longrightarrow V, \qquad G(a, x) = \alpha(a) + \pi(x),$$

uses the first ℓ coordinates as targets, J as helpers, and has message space V. Its helper kernel is K and the helper preimage of the target span is D, so its associated nested pair is $K \subseteq D$. $\square$

The second separation fixes the abstract nested pair itself. This is not a row-basis change of one inner code: by Proposition 3.2, such a change leaves (D_P, K_P) unchanged. Here a *coefficient presentation* means an ambient inner dual code, target coordinates, and target coefficient map that realize a prescribed abstract pair $K \subseteq D$ as its shortening–puncturing pair. Different such realizations may have different ambient dual codes and distances, so the examples below cannot be related by changing the row basis of one encoder. Let

$$K = 0, \qquad D = \langle u, v \rangle \le \mathbb{F}_2^3, \qquad u = 100, \quad v = 011.$$

The nonzero helper words have weights $1, 2, 3$, so $(M_1, M_2) = (1, 3)$.

Proposition 5.3 (The abstract nested pair does not determine confinement)**.** *Present the target space first by*

$$10 \mapsto u, \qquad 01 \mapsto v, \qquad 11 \mapsto u + v,$$

and then by

$$11 \mapsto u, \qquad 10 \mapsto v, \qquad 01 \mapsto u + v.$$

The two inner codes have the same associated nested pair, the same unlabelled set of helper supports, and the same complete relative-weight hierarchy. Their inner-dual weight multisets are respectively $\{2, 3, 5\}$ *and* $\{3, 3, 4\}$. *In every outer-distance regime where the collapse theorem applies, their additive costs are respectively*

$$(M_1 + d(I^\perp), M_2 + d(I^\perp)) = (3, 5) \qquad \textit{and} \qquad (M_1 + d(I^\perp), M_2 + d(I^\perp)) = (4, 6).$$

Proof. Use the graph-code realization

$$I^\perp = \{(-\alpha^{-1}(x), x) : x \in D\},$$

where α is the displayed identification from the two-dimensional target space to D. Over $\mathbb{F}_2$ the minus sign is invisible. The three nonzero graph words are

	target coefficient	helper word	target/helper weights	total weight
I_1	10	100	$1+1$	2
	01	011	$1+2$	3
	11	111	$2+3$	5
I_2	11	100	$2+1$	3
	10	011	$1+2$	3
	01	111	$1+3$	4

Thus $d(I^\perp)$ is two in the first presentation and three in the second. Adding this distance to $(M_1, M_2) = (1, 3)$ gives the two threshold sequences by Theorem 4.8. $\square$

These are limits on representation, not on executability. Keeping the functional labels avoids the second loss, and keeping minimizing lifts permits the exact recursion to return the original coefficient systems. The next section turns that closure law into an optimizer.

6 Exact recovery optimization

The closure law in Theorem 4.3 supplies an exact compiler state: its coordinates are the intermediate functionals that scalar summaries discard. ergodis (Exact Recovery, Global Optimization, and Invariant Synthesis) compiles and evaluates that state. Numerical tables store the fibrewise minima; separately stored predecessor labels and minimizing lifts reconstruct a coefficient-level witness for every optimum.

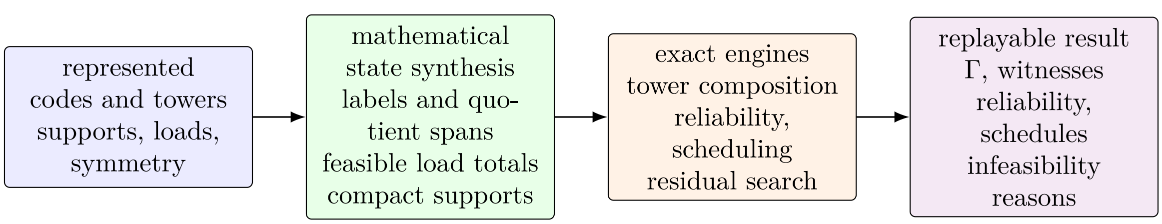

Figure 2: ergodis synthesizes the exact state exposed by the mathematics before search begins. Labelled coset costs drive tower composition, support families drive reliability, and feasible load totals are the states used in scheduling. Large support families may be stored by a zero-suppressed decision diagram (ZDD), which shares repeated choices. For attained fibres, stored lifts expand an optimum to a coefficient-level witness or obstruction; supported residual side constraints may be passed to CP-SAT. The mathematical results do not depend on executing this pipeline.

We first state the algorithm in flattened $\mathbb{F}_q$-linear coordinates. Let T have dimension t, let the outer message space have dimension m, and let the inner label space have dimension s. For the ith outer coordinate, write

$$A_i : \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^s) \longrightarrow \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^m)$$

for its contribution to the outer restriction map, and let $\lambda_i(b)$ be the prescribed-coset support cost of the inner label b. Different inner blocks may have different cost functions. For chosen block labels $b_1, \ldots, b_N$, the total outer label is $\sum_i A_i(b_i)$; the dynamic program enforces that this sum equals the requested label c.

Proposition 6.1 (Exact hierarchical recovery optimizer)**.** *For $c \in \mathrm{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^m)$, define*

$$D_0(c) = \begin{cases} 0, & c = 0, \\ +\infty, & c \neq 0, \end{cases} \qquad D_i(c) = \min_b \bigl(D_{i-1}(c - A_i b) + \lambda_i(b)\bigr). \tag{30}$$

Then $D_N(c)$ is the exact prescribed-coset support cost of the concatenated code at label c. Storing a minimizing label b and predecessor state at each finite entry returns an exact block-labelled recovery witness, not only its scalar cost. The same recursion applies to the target-normalized formula after adjoining its target-image labels to the state.

If every label is enumerated, put $Q = q^{mt}$ and $S = q^{st}$. The direct table algorithm uses $O(NQS)$ min–sum transitions, $O(Q)$ working cost memory, and $O(NQ)$ predecessor entries when all witnesses are retained.

Proof. After i blocks, $D_i(c)$ is the least sum of inner support costs among labels $(b_1, \ldots, b_i)$ satisfying $\sum_{j \leq i} A_j b_j = c$. This invariant is true at $i = 0$. Conditioning on the last label b_i gives exactly (30), so induction proves the claim. The leaf supports lie in disjoint blocks, hence their costs add. Backtracking the stored minimizers reconstructs the labels; the inner prescribed-coset witnesses then reconstruct the coefficient-level recovery system. There are Q states and S candidate labels per block, which gives the stated bounds. The target-normalized case is the same invariant applied to the paired labels in (15). ☐

The implementation also removes unreachable labels and merges states only when an explicit expansion map identifies each reduced choice with original block labels of the same cost. Backtracking through that map recovers an original-instance witness. Quotient coordinates, projectively equivalent columns, and repeated block types are instances of this implementation rule; they are not additional claims of Proposition 6.1.

The resulting table supplies all exact helper costs needed by the finite confinement formula, including its zero and nonzero outer-functional sectors. It can therefore emit a replayable recovery witness when a threshold fails, or a complete exhausted-state certificate when the requested label is unreachable. Iterating the compiler through a tower preserves the intermediate labels required at the next level; collapsing each table to one minimum would destroy precisely this compositional information.

The same support functions also lead to an exact capacity-aware repair scheduler. Suppose demand i has at most A admissible recovery choices. Choice a consumes a nonnegative integer load vector $\ell_{i,a} \in \mathbb{Z}_{\geq 0}^H$ on H helpers, and capacities are $\boldsymbol{c} = (c_1, \ldots, c_H)$. A demand may be skipped. This model distinguishes support occupancy from coefficient-weighted download and permits different capacities at different helpers.

Proposition 6.2 (Exact capacitated repair scheduling)**.** *Let $\mathcal{S} = \prod_{h=1}^H \{0, \ldots, c_h\}$. A dynamic program indexed by the current load $u \in \mathcal{S}$, with transitions $u \mapsto u + \ell_{i,a}$ whenever the result is* *within capacity and with a skip transition, returns the maximum number of simultaneously served demands and an exact assignment witness. It uses*

$$O\left(DAH\prod_{h=1}^{H}(c_h+1)\right)\quad \textit{time and}\quad O\left(\prod_{h=1}^{H}(c_h+1)\right)\quad \textit{working memory,} \tag{31}$$

apart from stored predecessors.

If there are positive integer weights w_h and a common $M > 0$ such that $w \cdot \ell_{i,a} = M$ for every admissible choice, a state serving exactly r demands lies on the graded shell

$$\mathcal{S}_r = \{u \in \mathcal{S} : w \cdot u = rM\}.$$

Thus the full box may be replaced by the union of its reachable shells. Two distinct states on one shell are incomparable in the componentwise order, so ordinary Pareto-dominance pruning cannot remove either one. The grading is therefore both an exact compression and a certificate of the limit of that common pruning rule.

For example, if every served repair adds unit total load, all states serving r demands lie on $\sum_h u_h = r$. At a fixed served count, decreasing one helper load necessarily increases another, so neither load vector dominates the other; objective-aware Pareto pruning cannot discard either state.

Proof. After processing the first i demands, associate to every reachable load the largest served count and a predecessor realizing it. The skip and assignment transitions enumerate exactly the feasible decisions for demand $i + 1$; induction gives optimality and witness recovery. Scanning at most A choices across H coordinates for every demand and box state gives (31). Under the grading hypothesis, additivity places every r-repair load on $\mathcal{S}_r$. If $u \le v$ componentwise and both lie on the same shell, positivity of w forces $u = v$, proving the antichain assertion. □

6.1 Measured performance and solver choice

The scientific role of the implementation is the exact executable semantics above. The measurements in this subsection assess engineering reach and the point at which a general solver becomes preferable; they are not mathematical evidence or the principal contribution of the software.

The specialized dynamic programs are most useful when compilation exposes a small algebraic state space. They are not replacements for a general constraint solver when the residual model contains unrelated side constraints. The appropriate deployment rule is therefore:

$$\text{algebraic compilation} \longrightarrow \begin{cases} \text{specialized exact DP,} & \text{when the compiled state is sufficient,} \\ \text{CP-SAT,} & \text{for the remaining general constraints.} \end{cases}$$

CP-SAT combines constraint programming, satisfiability, and linear-relaxation methods [35]; it is therefore an appropriate end-to-end baseline. A second, *structured CP-SAT* baseline receives the same symmetry reduction and algebraic preprocessing as ergodis, isolating the value of the specialized state engine from the value of compilation itself.

Measurement protocol. The measurements were made on an AMD Ryzen AI 9 HX 370 under Linux. Each paired program used the same selected logical CPU, and their order was rotated to limit ambient-load bias. Elapsed times are monotonic in-process times and include construction of the compiled state or control model. The application controls use eleven rounds and the

Rust kernels seven; reported values are medians of the raw samples. The only exceptions are explicitly identified single completed deterministic solves. Peak memory is the process high-water resident set size. Exact versions, checksums, raw samples, and replay commands are recorded in `ergodis/evidence/benchmarks.json`.

The cloud-LRC scheduling kernel in Table 1 operates on the exact six-type quotient of the 100,000 demands and is timed in a repeated inner loop; its displayed "$< 1\,\mu$s" value is therefore an amortized solve time for the preaggregated six-type instance, not a claim about parsing 100,000 input records. We use only the conservative ratio implied by that displayed upper bound.

Compiler versus residual engine. The benchmark suite separates the algebraic compiler from the residual state engine. On four exact capacitated-scheduling profiles, ergodis was 2.5–377 times faster than direct CP-SAT and 2.5–373 times faster than structured CP-SAT receiving the same safe preprocessing. The represented tower benchmark isolates the composition theorem: at depth six and fanout four, ergodis expands an exact coefficient witness over 4096 leaves in $766\,\mu$s, while direct and labelled-table CP-SAT take 264 s and 6.19 s, respectively. The direct entry is one completed deterministic solve; the shorter entries are interleaved medians.

The same compilation pattern has been tested through six bundled application examples. These are separate matched comparisons, not a ranking of unlike tasks. Each row below states the question and the exact output checked against its formulation-specific open-source control. The narrowest gap is eightfold; the structure-dominated examples reach three to five orders of magnitude. The comparisons do not include commercial solvers or claim a universal ranking.

example	question/output checked	bounded instance	exact control	ergodis	control	faster
recursive XOR	complete minimal-support family	8 diamonds, 256 supports	Graphillion ZDD [19]	$102\,\mu$s	$864\,\mu$s	$8\times$
cloud LRC	maximum served count and loads	100,000 demands, six types	HiGHS integer model	$< 1\,\mu$s	$4877\,\mu$s	$> 4877\times$
repair DAG	optimal makespan and schedule	3×21 tasks	CP-SAT intervals	$2\,\mu$s	$19{,}218\,\mu$s	$7881\times$
QC-LDPC	absence of a weight-four word	lift 50,000	CryptoMiniSat XOR	$1517\,\mu$s	$509{,}306\,\mu$s	$336\times$
vector repair	minimum nodes and spanning set	64 nodes, 2 symbols/node	CryptoMiniSat XOR	$10\,\mu$s	$48{,}691\,\mu$s	$4717\times$
GPU MDS	feasible flow and helper assignment	10,000 shards, 64 failures	OR-Tools max-flow	$100\,\mu$s	$103{,}061\,\mu$s	$1029\times$

Table 1: Six separate exact comparisons. ZDD means zero-suppressed decision diagram; LRC means locally repairable code; DAG means directed acyclic graph; QC-LDPC means quasi-cyclic low-density parity-check code; MDS means maximum-distance-separable code. The cloud-LRC ratio is the conservative bound supported by the displayed censored time; other ratios use unrounded medians.

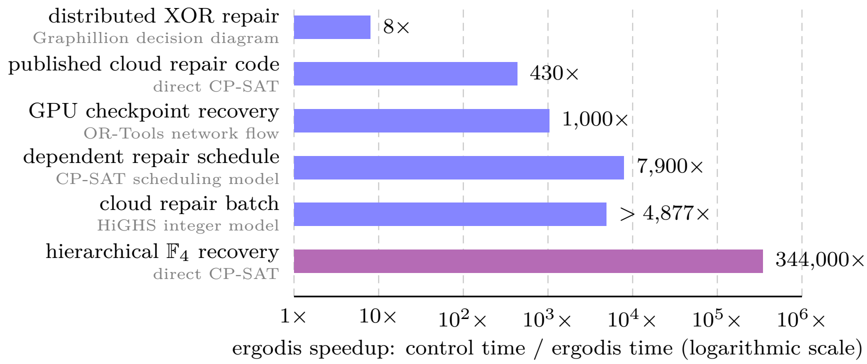

Figure 3: How many times faster ergodis is than matched exact controls for representative applications and the theorem-driven tower computation. Longer bars mean larger ergodis speedups. Each formulation-specific control is printed beneath its workload; full instances appear in Table 1, with the published cloud-code check below. The cloud-code row is Jin and Fu's family [20]; the cloud-batch bar uses the conservative lower bound implied by its censored time. The distinguished hierarchical $\mathbb{F}_4$ row compares labelled min–sum composition with direct CP-SAT and isolates the reduction supplied by the closure theorem. Controls differ by row, so this is not a general solver ranking.

As a code-native check, ergodis also analyzes Jin and Fu's published $\mathbb{F}_4$ Hamming-outer family [20, Corollary 5.4]. For its binary $[4095, 2718, 6; 2]$ LRC, all backends obtain the same exact confinement cost; the 21-run ergodis median is 231 ms, versus one completed optimality proof of 100 s for direct CP-SAT and 82 s for labelled-table CP-SAT. Exact inputs, outputs, tool versions, peak memory, replay commands, and raw timing samples are in `ergodis/evidence/benchmarks.json` and `ergodis/BENCHMARKS.md`; the two minute-scale CP-SAT entries are single completed deterministic solves.

General-purpose constraint solvers have also been used to construct binary codes [21], and repair dependency graphs and scheduling predate this work [23]. The contribution here is the structure-aware compilation of linear-recovery fibres, with exact witnesses, before a specialized dynamic program or a residual solver is invoked.

The largest gains arise when the algebraic reductions expose a low-dimensional reachable set inside a much larger raw model. ergodis solves that state directly when it is sufficient and otherwise supplies an exact preprocessor for a residual CP-SAT model, retaining recovery witnesses and confinement thresholds in either case.

The optimizer changes no mathematical hypothesis: it evaluates the labelled state proved sufficient above. We now return to consequences of the collapsed RGHW threshold, beginning with ordinary generalized weights and extremal inner codes.

7 Consequences of the threshold

The fixed-target cost refines ordinary generalized weights, which optimize over target sets, and is bounded above by worst-case cooperative locality, which maximizes over them. Let $d_e(C^\perp)$ denote the eth generalized Hamming weight of $C^\perp$ [40]. For an e-set $P \subseteq E$, let $\kappa_C(P)$ be the least $|H|$ for which some e-dimensional subcode $A \leq C^\perp$, supported on $P \cup H$, restricts isomorphically to $\mathbb{F}_q^P$, so every target coefficient vector has a unique equation in A. Set $\kappa_C(P) = \infty$ when no such subcode exists. Thus $\kappa_C(P)$ is the minimum number of helpers needed to recover all coordinates in P simultaneously.

The first identity below is the standard information-set interpretation of generalized Hamming weights; we record it to derive the paper-specific nonconfinement consequence.

Theorem 7.1 (Best-target generalized weight and first nonconfinement cost)**.** *For* $1 \le e \le \dim C^\perp$*,*

$$\min_{\substack{P \subseteq E \\ |P|=e}} \kappa_C(P) = d_e(C^\perp) - e. \tag{32}$$

Consequently, if every e*-set is recoverable and* $r_e^{\mathrm{coop}}(C) = \max_{|P|=e} \kappa_C(P)$*, then*

$$r_e^{\mathrm{coop}}(C) \ge d_e(C^\perp) - e.$$

For a fixed inner code C*, once the outer dual distance is large enough for the collapse in Theorem 4.7, the least helper cost of a first nonconfined system over all* e*-coordinate target sets is*

$$d_e(C^\perp) - e + d(C^\perp). \tag{33}$$

Proof. A normalized identity system on P spans an e-dimensional subcode of $C^\perp$ whose support is P together with its helper union. Hence $d_e(C^\perp) \le e + \kappa_C(P)$.

Conversely, let $A \le C^\perp$ have dimension e and support size $d_e(C^\perp)$. Choose e linearly independent columns of a generator matrix of A; their coordinate set P is an information set and has size e, and restriction $A \to \mathbb{F}_q^P$ is an isomorphism. Row reduction on these columns gives normalized identity equations whose helper union is $\mathrm{supp}(A) \setminus P$. This proves (32); the cooperative-locality inequality follows by taking the maximum instead of the minimum; compare [1, Theorem 1].

It remains to justify the nonconfinement formula when the target columns indexed by P are dependent. Put $U = \mathrm{im}\, G_P$. A normalized identity system on $\mathbb{F}_q^P$ and a normalized recovery system on U have the same minimum helper union. Indeed, restricting an identity system along a linear section $\alpha : U \to \mathbb{F}_q^P$ gives a system on U. Conversely, given a system on U, write

$$a = \alpha(G_P a) + \big(a - \alpha(G_P a)\big) \qquad (a \in \mathbb{F}_q^P).$$

Thus $G_P : \mathbb{F}_q^P \twoheadrightarrow U$ separates each target vector into its chosen lift $\alpha(G_P a)$ and a kernel relation. The second summand lies in $\ker G_P$ and hence is a target-only dual equation, so adjoining these kernel equations recovers the full identity without adding helpers. Thus the common minimum is $\kappa_C(P)$.

Now repeat the block-functional argument of Theorem 4.7. Once the outer-functional case is excluded, the target block of any identity system uses at least $\kappa_C(P)$ helpers. If the system is nonconfined, a nonzero equation map in another block has union support at least $d(C^\perp)$; the two supports are disjoint. Conversely, attach a minimum-weight word of $C^\perp$ in a second block through any nonzero linear functional on $\mathbb{F}_q^P$. Hence the first nonconfinement cost for this P is $\kappa_C(P) + d(C^\perp)$. Minimizing over P and using (32) proves (33). $\square$

Before bounding the MDS thresholds, we record what the coefficient layer adds even when the support clutter is completely uniform.

Theorem 7.2 (Minimum-weight MDS recovery equations span the dual)**.** *Let* $C \le \mathbb{F}_q^E$ *be an* $[n,k]_q$ *MDS code with* $1 \le k < n$*, and let* $x \in E$*. Then*

$$\mathrm{span}\, \mathcal{P}_x^{(\le k)}(C) = C^\perp,$$

and their supports form the complete k*-uniform clutter on* $E \setminus \{x\}$*.*

Equivalently, the minimum-weight dual equations through x realize every k-subset of the other coordinates as a helper support.

Proof. The dual has dimension $n-k$ and distance $k+1$. For each $(k+1)$-set T, restriction $C^\perp \to \mathbb{F}_q^{E\setminus T}$ has a nonzero kernel, whose words have support exactly T by the distance bound. Thus all stated supports occur. Fix a $(k-1)$-set $Q \subseteq E \setminus \{x\}$ and take the normalized word on $\{x,t\}\cup Q$ for each $t \in E\setminus(\{x\}\cup Q)$. Their private coordinates t make these $n-k$ words independent, hence a basis of $C^\perp$. $\square$

The support clutter depends only on the MDS parameters, whereas the normalized equations retain the represented dual code.

Put $k = \dim I$, $u = \operatorname{rank} G_P$, and $b = \operatorname{rank} G_J$. The exact sequence and the relative and ordinary Singleton bounds give

$$M_t(D_P, K_P) \le b - \ell + t, \qquad d(I^\perp) \le k+1. \tag{34}$$

The three ranks governing the statement are

$$\begin{array}{r|l} u = \operatorname{rank} G_P & \text{dimension visible on the target coordinates} \\ b = \operatorname{rank} G_J & \text{dimension visible on the helper coordinates} \\ \ell = \dim(\operatorname{im} G_P \cap \operatorname{im} G_J) & \text{dimension recoverable from the helpers.} \end{array}$$

Theorem 7.3 (MDS ceiling and rank-one rigidity)**.** Universal ceiling. *For $1 \le t \le \ell$,*

$$M_t(D_P, K_P) + d(I^\perp) \le 2k - \ell + t + 1. \tag{35}$$

MDS staircase. *Suppose I is MDS. Then*

$$u = \min\{k, |P|\}, \qquad b = \min\{k, |J|\}.$$

Then $\ell = u + b - k$. If $\ell \ge 1$, then, for $1 \le t \le \ell$,

$$M_t(D_P, K_P) = k - u + t, \qquad M_t(D_P, K_P) + d(I^\perp) = 2k - u + t + 1. \tag{36}$$

If $|J| \ge k$, then $\ell = u$ and these values attain the ceiling in (35) at every rank. Rank-one rigidity. *Conversely, when $\ell \ge 1$, equality in (35) at $t = 1$ forces I to be MDS and forces equality at every rank.*

Proof. The ceiling follows from (34) and $b \le k$. For an MDS code every set of at most k generator columns is independent; equivalently, the column matroid is uniform. Thus $\operatorname{rank} G_P = u$, $\operatorname{rank} G_J = b$, $\ell = u + b - k$, and, for $H \subseteq J$ with $|H| = h \le k$,

$$\dim\bigl(\operatorname{span} G_P \cap \operatorname{span} G_H\bigr) = \max\{0, u + h - k\}.$$

This is the usual dimension count for two subspaces generated by disjoint subsets of a uniform k-dimensional column configuration. Thus Proposition 3.4 gives $M_t = k - u + t$ for $1 \le t \le \ell$; here $k - u + t \le b \le |J|$. Now $d(I^\perp) = k+1$, and if $|J| \ge k$, then $b = k$ and $\ell = u$, proving the remaining direct assertions.

Equality at $t = 1$ forces equality in both Singleton bounds and $b \le k$. Hence $d(I^\perp) = k+1$; equality in the dual Singleton bound is precisely the MDS condition for I. Also $M_1 = b - \ell + 1$. Strict growth then gives $M_t \ge b - \ell + t$, while relative Singleton gives the reverse inequality; every rank is extremal. $\square$

Thus extremality is detected at rank one: equality in the first threshold forces the inner code to be MDS and forces every later rank to attain its ceiling. No separate higher-rank hypothesis is needed.

We next show that the transferred local structure can occupy a positive fraction of an asymptotically good concatenated family, rather than appearing only in one distinguished block. Let the inner code have parameters $[m, k, d]_q$, so $[L : \mathbb{F}_q] = k$. Let $K_N = \dim_L O_N$ and $\Delta_N = d(O_N)$.

Corollary 7.4 (Positive-density realization). *Assume $d(O_N^{\perp}) \to \infty$. Fix either a nonzero target subspace T and a radius r satisfying Theorem 4.7, or a rank t and radius r satisfying Theorem 4.8. For all sufficiently large N, every corresponding inner normalized recovery system of cost at most r and its exact helper support are copied in each of the N blocks. Each fixed inner coordinate type has density $1/m$ among the concatenated coordinates; the union of the types in P has density $|P|/m$.*

In particular, if $r < M_1(D_P, K_P) + d(I^{\perp})$, then all cost-$\leq r$ systems for all nonzero subspaces of W_P are copied simultaneously in every block by Corollary 4.6. Hence the complete bounded local recovery data through radius r occur on the stated positive-density coordinate classes.

The concatenated code has length mN, dimension kK_N, and minimum distance at least $d\Delta_N$. If

$$\frac{K_N}{N} \longrightarrow R > 0, \qquad \liminf_N \frac{\Delta_N}{N} \geq \delta > 0,$$

then the concatenated family has rate kR/m and relative distance at least $d\delta/m$. Outer families with positive limiting rate, positive primal relative distance, and dual distance tending to infinity exist over every finite field L.

Proof. The transfer statement is the zero-extension bijection in the confinement theorems, applied independently in every block. There are mN coordinates, and each inner coordinate occurs once in each block, which gives the density claims. The blockwise inner encoder is injective, so restriction of scalars gives dimension kK_N. A nonzero outer word has at least Δ_N nonzero symbols, each encoding to an inner word of weight at least d. This proves the distance and asymptotic assertions.

For the existence statement, write $Q = |L|$. Choose $0 < R < 1$ and positive $\delta, \delta^{\perp}$ such that

$$H_Q(\delta) < 1 - R, \qquad H_Q(\delta^{\perp}) < R,$$

where H_Q is the standard Q-ary entropy function measuring the exponential growth rate of a Q-ary Hamming ball. The first-moment argument for a random K_N-dimensional L-linear code bounds the expected numbers of nonzero primal and dual words below the two cutoffs, up to subexponential factors, by

$$Q^{N(H_Q(\delta)+R-1)} \qquad \text{and} \qquad Q^{N(H_Q(\delta^{\perp})-R)},$$

respectively. Both tend to zero by the displayed inequalities, so one may choose a sequence with rate tending to R, primal relative distance at least δ, and dual relative distance at least $\delta^{\perp}$; see also [27, Chapter 17]. ☐

Exact support transfer also preserves the fractional load-balancing region used in service-rate problems. We state the consequence because it requires only the inclusion-minimal recovery sets, whereas the theorem transfers all normalized coefficients as well.

Fix one inner target block with helper set J. Let $\mathcal{A}$ be a finite set of demands on that block, with demand a represented by a nonzero target subspace $T_a \leq W_P$, and let $\mathcal{M}_a^{(\leq r)}$ be the inclusion-minimal exact helper supports of size at most r for $a \in \mathcal{A}$. For a nonnegative helper-capacity vector

c, define

$$\Lambda_{\mathcal{A}}^{(\le r)}(c) = \left\{ \lambda \ge 0 : \begin{array}{l} \text{there are } f_{a,R} \ge 0 \text{ with } \lambda_a = \sum_{R \in \mathcal{M}_a^{(\le r)}} f_{a,R}, \\ \displaystyle\sum_a \sum_{R \ni j} f_{a,R} \le c_j \text{ for every helper } j \end{array} \right\}.$$

This is the bounded service-rate region of the recovery-set system [3, 4, 30, 9].

Corollary 7.5 (Transfer of bounded service-rate regions)**.** *Let $O \le L^N$ be an outer code with $N \ge 2$ whose projection onto the target block is nonzero. Suppose*

$$r < \Gamma_{j,T_a}(O, I) \qquad (a \in \mathcal{A}).$$

Transport capacities from the inner helper set to its copy in that block of $O \circ I$. Then the inner and concatenated bounded service-rate regions are equal. Capacities assigned outside the target block do not change this equality, because confinement excludes every cost-$\le r$ recovery using those external helpers. In particular, the hypothesis follows from $d(O^\perp) > r + 1$ together with $r < \rho_{T_a}(I) + d(I^\perp)$ for every $a \in \mathcal{A}$. Consequently, for an outer family with $d(O_N^\perp) \to \infty$, the equality holds for all sufficiently large N whenever those inner inequalities hold.

Proof. Theorem 4.1 confines every cost-$\le r$ system for every demand. Thus restriction and zero-extension copy every inclusion-minimal support and create no cross-block minimal support. The helper bijection therefore transports the feasible variables $f_{a,R}$ and every capacity inequality in both directions. Allowing the full upward-closed recovery-set family gives the same region: flow assigned to a nonminimal set can be moved to a contained minimal set and weakly decreases every helper load. Thus cross-block supersets and capacities outside the target block are irrelevant. □

The next family realizes all three information levels in closed form: its RGHWs give minimum recovery costs, its ambient dual distance gives the confinement thresholds under the outer-distance condition, and its projective rank distribution gives reliability beyond the RGHW hierarchy.

8 The projective-simplex family

This non-MDS family realizes the paper's information hierarchy in one model. Its relative weights determine the minimum costs, its ambient inner dual gives the collapsed confinement thresholds, and the full projective rank distribution determines reliability. Thus the family shows both what the RGHW hierarchy captures and what it forgets. Assume $m \ge 2$, put

$$N = \frac{q^m - 1}{q - 1},$$

and let A be an $m \times N$ matrix containing one representative of every point of $\mathrm{PG}(m-1, q)$. Here $\mathrm{PG}(m-1, q)$ is the set of one-dimensional subspaces of $\mathbb{F}_q^m$, and A chooses one nonzero vector from each such subspace. Define the inner code by

$$I_m^\perp = \{(a, aA) : a \in \mathbb{F}_q^m\}. \tag{37}$$

Take the first m coordinates as targets and the projective coordinates as helpers. The associated nested pair is isomorphic to

$$0 \subseteq S_m = \{aA : a \in \mathbb{F}_q^m\},$$

where S_m is the q-ary simplex code. The generalized-weight formula for S_m is classical; in fact every t-dimensional simplex subcode has the same support size [34]. We include the short projective count to connect that hierarchy to the confinement and reliability statements below.

Theorem 8.1 (Projective-simplex thresholds)**.** *For $1 \le t \le m$,*

$$M_t(S_m, 0) = \frac{q^m - q^{m-t}}{q-1}, \tag{38}$$

and

$$d(I_m^\perp) = q^{m-1} + 1. \tag{39}$$

Hence

$$C_t := M_t(S_m, 0) + d(I_m^\perp) = \frac{q^m - q^{m-t}}{q-1} + q^{m-1} + 1. \tag{40}$$

In outer families with dual distance tending to infinity, the collapsed threshold C_t equals $\Gamma_{j,t}(O, I_m)$ once the outer dual distance reaches $C_t + 1$. Equivalently, at every fixed helper radius the corresponding confinement statement holds for all sufficiently large members of the family. The family is non-MDS for $m \ge 2$; for $m \ge 3$ the successive increments of C_t are unequal.

Proof. The target restriction of a word $(a, aA) \in I_m^\perp$ is a, and no nonzero dual word has zero target restriction. Hence normalized recovery systems correspond, with support preserved, to subcodes of S_m; this proves the asserted identification of the associated pair.

Let $U \le \mathbb{F}_q^m$ have dimension t. The common zero coordinates of the subcode UA are the projective points in $U^\perp$, a subspace of projective dimension $m - t - 1$. Thus

$$|\operatorname{supp}(UA)| = N - \frac{q^{m-t} - 1}{q-1} = \frac{q^m - q^{m-t}}{q-1},$$

independently of U. This proves (38). Every nonzero simplex word aA has weight q^{m-1}, while a nonzero target vector a has weight at least one. Equality is attained by a coordinate vector, which proves (39). The confinement formula is Theorem 4.8. The non-MDS and increment assertions follow from the displayed parameters. □

question	retained information	exact answer
minimum rank-t helper union	RGHW hierarchy	$\frac{q^m - q^{m-t}}{q-1}$
eventual first nonconfined cost	RGHW plus $d(I_m^\perp)$	$\frac{q^m - q^{m-t}}{q-1} + q^{m-1} + 1$
rank-t recovery reliability	projective rank distribution	$R_t(s)$ in (42)

Table 2: The projective-simplex family separates the three questions. The first two use minimum-cost data; the stochastic law requires the distribution of ranks of failed projective sets.

This family also gives an exact stochastic law at every recovered dimension. Let each helper survive independently with probability s, and let F be the set of failed projective points. The available recovery equations are indexed by

$$\{a \in \mathbb{F}_q^m : aA|_F = 0\} = \big(\operatorname{span} F\big)^\perp,$$

which has dimension $m - \operatorname{rank}(F)$. Write $\begin{bmatrix} a \\ b \end{bmatrix}_q$ for the Gaussian binomial coefficient, the number of b-dimensional vector subspaces of $\mathbb{F}_q^a$. A projective r-frame here means r linearly independent projective points, or equivalently the projectivization of a vector-space basis of its r-dimensional span.

Theorem 8.2 (Projective rank reliability)**.** Exact probability. *The probability of retaining at least t independent target recovery equations is*

$$R_t(s) = \sum_{\substack{F \subseteq \mathrm{PG}(m-1,q) \\ \mathrm{rank}(F) \leq m-t}} (1-s)^{|F|} s^{N-|F|}. \tag{41}$$

Closed form. *Writing $N_v = (q^v - 1)/(q-1)$, this equals*

$$R_t(s) = \sum_{u=0}^{m-t} \begin{bmatrix} m \\ u \end{bmatrix}_q \sum_{v=0}^{u} \begin{bmatrix} u \\ v \end{bmatrix}_q (-1)^{u-v} q^{\binom{u-v}{2}} s^{N-N_v}. \tag{42}$$

Endpoint behavior. *Its endpoint terms are*

$$R_t(s) = \begin{bmatrix} m \\ t \end{bmatrix}_q s^{M_t} + O(s^{M_t+1}) \qquad (s \to 0), \tag{43}$$

$$1 - R_t(s) = B_{m,m-t+1}(1-s)^{m-t+1} + O((1-s)^{m-t+2}) \qquad (s \to 1), \tag{44}$$

where

$$B_{m,r} = \begin{bmatrix} m \\ r \end{bmatrix}_q \frac{|\,\mathrm{GL}(r,q)|}{r!(q-1)^r}$$

is the number of unordered projective r-frames.

The first endpoint says that rare successful recovery is dominated by the $\left[\begin{smallmatrix} m \\ t \end{smallmatrix}\right]_q$ minimum helper supports of size M_t. The second says that near-perfect availability first fails on a smallest projective frame whose rank exceeds the allowed failure span.

Proof. Recovery of t independent target combinations is possible exactly when $m - \mathrm{rank}(F) \geq t$, which gives (41). For a vector subspace $U \leq \mathbb{F}_q^m$, let $\mathbb{P}(U)$ denote its projective points. Möbius inversion in the subspace lattice gives the total probability of failed sets spanning a fixed u-space U as

$$\sum_{V \leq U} \mu(V,U) s^{N-N_{\dim V}}, \qquad \mu(V,U) = (-1)^{u-v} q^{\binom{u-v}{2}}, \qquad v = \dim V.$$

Here the total weight of all failed sets contained in $\mathbb{P}(V)$ is

$$\sum_{F \subseteq \mathbb{P}(V)} (1-s)^{|F|} s^{N-|F|} = s^{N-N_{\dim V}},$$

which accounts for the power of s before inversion. There are $\left[\begin{smallmatrix} m \\ u \end{smallmatrix}\right]_q$ choices for U and $\left[\begin{smallmatrix} u \\ v \end{smallmatrix}\right]_q$ subspaces $V \leq U$ of dimension v. Summing over $u \leq m-t$ proves (42).

For $s \to 0$, a smallest surviving set that permits rank-t recovery is the complement of the points in an $(m-t)$-dimensional vector subspace. It has size M_t, and there are $\left[\begin{smallmatrix} m \\ m-t \end{smallmatrix}\right]_q = \left[\begin{smallmatrix} m \\ t \end{smallmatrix}\right]_q$ such subspaces. For $s \to 1$, failure first occurs when the failed points span dimension $m-t+1$. A smallest such set is a projective frame of that size. Counting its span and then its unordered projective bases gives $B_{m,m-t+1}$. ☐

At the same helper length, the simplex presentation is characterized by its first recovery cost: equality means that minimum cost is q^{m-1}. This is the length-N, multiplicity-one specialization of Bonisoli's classification of linear equidistant codes, whose nonzero words all have the same weight, [6]; the proof below records the precise zero-column and projective-multiplicity conventions used here.

Proposition 8.3 (Equality in the first projective bound)**.** *Let B be any full-row-rank $m \times N$ matrix over $\mathbb{F}_q$, with $N = (q^m - 1)/(q-1)$. Then*

$$\min_{0 \neq a \in \mathbb{F}_q^m} \mathrm{wt}(aB) \leq q^{m-1}.$$

Equality holds if and only if, up to column scaling and permutation, the columns of B represent every point of $\mathrm{PG}(m-1, q)$ *exactly once.*

Proof. For each nonzero column, the fraction of nonzero $a \in \mathbb{F}_q^m$ for which its entry in aB is nonzero is

$$\frac{q^{m-1}(q-1)}{q^m - 1}.$$

Averaging $\mathrm{wt}(aB)$ over $a \neq 0$ gives at most q^{m-1}, strictly less if B has a zero column. If the minimum attains this average, every nonzero linear combination has weight q^{m-1} and every projective hyperplane contains total column multiplicity $(q^{m-1} - 1)/(q-1)$.

Let c be the vector of projective column multiplicities. The point–hyperplane incidence matrix is square, and its real Gram matrix has the form $(r - \lambda)I + \lambda J$: here r is the number of hyperplanes through one point and λ the number through two distinct points, with $r > \lambda$. It is nonsingular, so the constant hyperplane-sum equations have the unique solution $c = \mathbf{1}$. Conversely, the simplex matrix has constant nonzero weight q^{m-1}. □

9 Verification scope

The paper has a paper-owned Lean 4 companion built against a pinned Mathlib revision. It formalizes the construction of the associated nested code pair from abstract target and helper generator maps. In the notation of (1), the checked statements are

$$K_P \leq D_P, \qquad G_J(D_P) = W_P, \qquad G_J|_{D_P} : D_P \twoheadrightarrow W_P, \qquad \ker(G_J|_{D_P}) = K_P.$$

The corresponding declarations are

Paper statement	Lean declaration
$K_P \leq D_P$	`helper_ker_le_helperCodeForTargetSpace`
$G_J(D_P) = W_P$	`map_helperCodeForTargetSpace_eq_recoverableTargetMessageSpace`
surjectivity	`recoverableTargetMap_surjective`
restricted kernel	`mem_ker_recoverableTargetMap_iff`

The companion uses Lean 4.32.0-rc1 and the Mathlib commit

`571b8a8e54219b4d393f75f4b8653fac08197fcc`.

The axiom audit of all four reviewer-facing declarations reports only

{`Classical.choice`, `Quot.sound`, `propext`}.

These are routine logical and quotient principles used throughout Mathlib, not unproved assumptions about codes or recovery. No coding-theory result is introduced as a Lean axiom.

The formal coverage stops at the exact sequence. In particular, Proposition 3.2, Theorems 3.3, 4.1, 4.7, 4.8, and 4.9, as well as Proposition 4.3, the strictness construction in Proposition 4.10 and the later consequences and examples, are human-proved and are not established by the paper-local

Lean companion. The claim manifest records them as absent rather than treating the exact-sequence formalization as evidence for them.

The reliability polynomials in Proposition 5.1 follow from the displayed union-size table by inclusion–exclusion. No exhaustive computation is a premise of that proposition or of any theorem in the main proof chain. Machine-readable examples and replay checks may therefore be used to audit arithmetic without changing the logical dependence of the paper.

10 Conclusion

The central question is compositional: which recovery information must be retained before an inner code is placed inside an outer one? The canonical shortening–puncturing pair answers the single-block minimum-cost question. Its relative generalized Hamming weights are exactly the minimum helper unions for recovered subspaces of each dimension. They are not, however, sufficient recursive state for finite concatenation.

Exact finite transfer is captured by the prescribed-coset support functions indexed by the intermediate functional each block must realize. Minimizing their blockwise costs over all compatible outer labels gives $\Gamma_{j,T}(O, I)$, the minimum cost of a normalized recovery that leaves the target block. The same labelled costs compose by min–sum through finite towers. The numerical table stores one minimum per label; storing a backpointer as well recovers an optimizing coefficient-level witness. At rank one, tests by outer dual lines distinguish exactly what any outer code can observe. Moreover, any higher-rank nonconfinement witness contains a rank-one nonconfinement witness of no larger cost. Thus rank-one confinement controls every recoverable rank, and below that threshold restriction and zero-extension preserve all bounded coefficient and support families and the reliability or scheduling laws defined from them.

The additive threshold

$$M_t(D_P, K_P) + d(I^{\perp})$$

is the zero-functional sector of the exact formula. Outer dual distance makes it exact by excluding every nonzero-functional sector below the chosen radius. Restriction and zero-extension then preserve normalized equations and exact helper supports, not only a locality value. This coefficient-level bijection yields the positive-density and bounded service-rate transfers; the one-coordinate weighted formula can remain exact even when the ordinary outer-distance condition fails.

The separations identify the cost of each projection. The complete RGHW hierarchy determines minimum helper unions but not the overlap law governing bounded reliability. Even the binary image code, its dual, the abstract nested pair, the complete relative-weight hierarchy, and the unlabelled helper supports can agree while one fixed outer code distinguishes two extension-field presentations through their functional labels. A separate ambient-presentation example shows that $d(I^{\perp})$ supplies another degree of freedom invisible to the abstract pair. The projective-simplex family exhibits all three levels explicitly: RGHWs give its minimum costs, ambient dual distance gives its collapsed thresholds, and the projective rank distribution gives its reliability polynomial.

ergodis evaluates the closure theorem rather than serving as evidence for it. Its min–sum compiler stores labelled minima and predecessor lifts, returning exact hierarchical costs and replayable witnesses; its scheduler then optimizes the resulting recovery choices under heterogeneous capacities. The recorded matched exact-control comparisons measure this tradeoff on declared instances and do not enter any proof.

Two extensions require different state. Disjoint simultaneous service is an integral packing problem rather than a minimum-union or fractional service-rate problem. Bandwidth-aware array-code repair lets each helper transmit a subspace of functionals, so Hamming support is no longer the relevant cost. Neither extension follows by relabelling the present invariant; each requires a new compositional cost model.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

References

[1] Khaled A. S. Abdel-Ghaffar and Jos H. Weber. Bounds for cooperative locality using generalized Hamming weights. In *2017 IEEE International Symposium on Information Theory*, pages 699–703, 2017.

[2] Srinivas M. Aji and Robert J. McEliece. The generalized distributive law. *IEEE Transactions on Information Theory*, 46(2):325–343, 2000.

[3] Mehmet Aktaş, Gauri Joshi, Swanand Kadhe, Fatemeh Kazemi, and Emina Soljanin. Service rate region: A new aspect of coded distributed system design. *IEEE Transactions on Information Theory*, 2021.

[4] Gianira N. Alfarano, Altan B. Kılıç, Alberto Ravagnani, and Emina Soljanin. The service rate region polytope. *SIAM Journal on Applied Algebra and Geometry*, 2024.

[5] Ferdinand Blomqvist, Oliver W. Gnilke, and Marcus Greferath. On decoding of generalized concatenated codes and matrix-product codes, 2020.

[6] Arrigo Bonisoli. Every equidistant linear code is a sequence of dual Hamming codes. *Ars Combinatoria*, 18:181–186, 1984.

[7] Yeow Meng Chee, Tuvi Etzion, Han Mao Kiah, and Hui Zhang. Recovery sets of subspaces from a simplex code. *IEEE Transactions on Information Theory*, 70(10):6961–6973, 2024.

[8] Hao Chen, San Ling, and Chaoping Xing. Quantum codes from concatenated algebraic-geometric codes. *IEEE Transactions on Information Theory*, 51(8):2915–2920, 2005.

[9] Priyanka Choudhary, Monika Yadav, and Maheshanand Bhaintwal. Maximal achievable service rates of some classes of linear codes, 2026.

[10] Dor Elimelech, Marcelo Firer, and Moshe Schwartz. The generalized covering radii of linear codes. *IEEE Transactions on Information Theory*, 67(12):8070–8085, 2021.

[11] Jr. G. David Forney. Generalized minimum distance decoding. *IEEE Transactions on Information Theory*, 12(2):125–131, 1966.

[12] Carlos Galindo, Fernando Hernando, Carlos Munuera, and Diego Ruano. Locally recoverable codes from the matrix-product construction, 2023. arXiv version 1; the identifier was subsequently replaced.

[13] Olav Geil. Considerate ramp secret sharing. *Designs, Codes and Cryptography*, 2026.

[14] Olav Geil, Stefano Martin, Ryutaroh Matsumoto, Diego Ruano, and Yuan Luo. Relative generalized Hamming weights of one-point algebraic geometric codes, 2014.

[15] Parikshit Gopalan, Cheng Huang, Huseyin Simitci, and Sergey Yekhanin. On the locality of codeword symbols. *IEEE Transactions on Information Theory*, 58(11):6925–6934, 2012.

[16] Anina Gruica, Benjamin Jany, and Alberto Ravagnani. LRCs: Duality, LP bounds, and field size. *Designs, Codes and Cryptography*, 94:100, 2026.

[17] Venkatesan Guruswami and Madhu Sudan. Decoding concatenated codes using soft information. In *Proceedings of the 17th Annual IEEE Conference on Computational Complexity*, pages 148–157, 2002.

[18] Venkatesan Guruswami and Mary Wootters. Repairing Reed–Solomon codes. *IEEE Transactions on Information Theory*, 63(9):5684–5698, 2017.

[19] Takeru Inoue, Hiroaki Iwashita, Jun Kawahara, and Shin ichi Minato. Graphillion: Software library for very large sets of labeled graphs. *International Journal on Software Tools for Technology Transfer*, 18:57–66, 2016.

[20] Hengfeng Jin and Fang-Wei Fu. Constructions of locally repairable codes via concatenated codes, 2026. arXiv:2605.04618v1.

[21] Stepan Kochemazov, Oleg Zaikin, Grigorii Trofimiuk, Kirill Antonov, and Alexander Semenov. Using constraint solvers to construct binary codes with good error correction performance. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 2026.

[22] Jun Kurihara, Tomohiko Uyematsu, and Ryutaroh Matsumoto. Secret sharing schemes based on linear codes can be precisely characterized by the relative generalized hamming weight. *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, E95-A(11):2067–2075, 2012.

[23] Shiyao Lin, Guowen Gong, Zhirong Shen, Patrick P. C. Lee, and Jiwu Shu. Boosting full-node repair in erasure-coded storage. In *2021 USENIX Annual Technical Conference*, pages 641–655. USENIX Association, 2021.

[24] Shu Liu, Liming Ma, Tingyi Wu, and Chaoping Xing. Good locally repairable codes via propagation rules, 2022. arXiv:2208.04484.

[25] Pavel Loskot and Norman C. Beaulieu. The input–output weight enumeration of binary Hamming codes. *European Transactions on Telecommunications*, 17(4):483–488, 2006.

[26] Yuan Luo, Chaichana Mitrpant, A. J. Han Vinck, and Kefei Chen. Some new characters on the wire-tap channel of type II. *IEEE Transactions on Information Theory*, 51(3):1222–1229, 2005.

[27] Florence Jessie MacWilliams and Neil James Alexander Sloane. *The Theory of Error-Correcting Codes*, volume 16 of *North-Holland Mathematical Library*. North-Holland, Amsterdam, 1977.

[28] Irene Márquez-Corbella, Edgar Martínez-Moro, and Carlos Munuera. Computing sharp recovery structures for locally recoverable codes, 2019.

[29] Gretchen L. Matthews, Griffin Matthews, and Eleanor Norton. Linear exact repair for dual decomposable codes. *Advances in Mathematics of Communications*, 2026.

[30] Arya Mazumdar. Capacity of locally recoverable codes. *IEEE Journal on Selected Areas in Information Theory*, 4:276–285, 2023.

[31] Mehryar Mohri. Weighted automata algorithms. In Manfred Droste, Werner Kuich, and Heiko Vogler, editors, *Handbook of Weighted Automata*, pages 213–254. Springer, 2009.

[32] Dmitry Yu. Nogin. Weight functions and generalized Hamming weights of linear codes. *Problems of Information Transmission*, 41(2):91–104, 2005.

[33] Lluis Pamies-Juarez, Henk D. L. Hollmann, and Frédérique Oggier. Locally repairable codes with multiple repair alternatives. In *2013 IEEE International Symposium on Information Theory*, pages 892–896, 2013.

[34] Xu Pan, Hao Chen, Hongwei Liu, and Shengwei Liu. Optimal few-GHW linear codes and their subcode support weight distributions, 2024.

[35] Laurent Perron, Frédéric Didier, and Steven Gay. The CP-SAT-LP solver. In *29th International Conference on Principles and Practice of Constraint Programming*, volume 280 of *LIPIcs*, pages 3:1–3:2, 2023.

[36] Ankit Singh Rawat, Arya Mazumdar, and Sriram Vishwanath. Cooperative local repair in distributed storage. *EURASIP Journal on Advances in Signal Processing*, 2015(1):1–17, 2015.

[37] Rodrigo San-José. An algorithm for computing generalized Hamming weights and the Sage package GHWs, 2025.

[38] Nihar B. Shah, K. V. Rashmi, P. Vijay Kumar, and Kannan Ramchandran. Distributed storage codes with repair-by-transfer and non-achievability of interior points on the storage–bandwidth tradeoff. *IEEE Transactions on Information Theory*, 58(3):1837–1852, 2012.

[39] James Singer. A theorem in finite projective geometry and some applications to number theory. *Transactions of the American Mathematical Society*, 43(3):377–385, 1938.

[40] Victor K. Wei. Generalized Hamming weights for linear codes. *IEEE Transactions on Information Theory*, 37(5):1412–1418, 1991.

[41] Min Ye and Alexander Barg. Explicit constructions of high-rate MDS array codes with optimal repair bandwidth. *IEEE Transactions on Information Theory*, 63(4):2001–2014, 2017.

[42] Yiwei Zhang, Eitan Yaakobi, Tuvi Etzion, and Moshe Schwartz. On the access complexity of PIR schemes. In *2019 IEEE International Symposium on Information Theory*, pages 2134–2138, 2019.